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Dynamic analysis of a spinning visco-elastic FG graphene platelets reinforced nanocomposite sandwich cylindrical shell with MRE core

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Abstract This paper investigates the dynamic responses of spinning FG GPLs reinforced with a nanocomposite sandwich cylindrical shell based on a magnetorheological elastomer (MRE) core subjected to thermomechanical loading and residual stress. The sandwich cylindrical shell is considered using the Donnell–Moshtari theory based on metal matrix nanocomposites; furthermore, GPLs are used with uniform and FG distribution in the thickness direction to reinforce these layers. The MRE core layer is modeled based on FSDT. The effect of temperature on the mechanical properties of MRE, GPLs, and metal matrix nanocomposites is considered. The mechanical properties of the nanocomposite sandwich shell are obtained based on the Halpin–Tsai micromechanics model and the rule of mixture. The equations of motion for a spinning sandwich shell are obtained by considering the rotary inertia and shear effect. The frequencies of a spinning shell are derived using the Differential Quadrature Method. The effect of parameters such as weight fraction and distribution of GPLs, MRE core, spinning speed, residual stresses, and thermomechanical loading on the dynamic behavior of spinning nanocomposite sandwich shells are studied.

1 Introduction

In most Engineering fields such as aerospace, automobile, and medicine, vibration control and stability of rotary shafts are highly important. Rotary shafts could be used as a boring tool holder for drilling parts [1] and can sometimes be the spinning shaft of a helicopter or airplane [2], the shaft of a gas turbine, or the ship's engine shaft [3]. Therefore, what is essential in these spinning structures is to prevent them from reaching critical speeds and protect them from destructive phenomena such as flutter [4]. To achieve the optimum design of spinning shafts, parameters affecting their vibration behavior should be fully recognized. The sandwich structure is one of the most widely utilized rotary structures in various industries today. The designer's primary goal in employing sandwich structures is to achieve an efficient design that uses each component for the best performance [5–7]. One of the ways to control the vibrations of rotating structures is to use magnetic fields and smart materials [8, 9]. Much research has been performed to optimize and strengthen sandwich structures and investigate various parameters affecting their vibrational behavior [10]. These researches include reinforcing such as graphene platelets [10–15] and carbon nanotubes and nanofibers [16–21]. The main properties of graphene plates are light, electric, and thermal conductivity, sufficient mechanical stiffness, and surface structure. Graphene platelets could be used as mechanical strengthening agents in polymeric or metallic matrices [22]. The graphene platelets are less expensive than nanotubes and nanofibers and can enhance the thermal conductivity of the structure by forming thermal lattices in composites [23–26]. Based on HSDT, Qaderi et al. [27] investigated the vibration of graphene platelet-reinforced composite plate under thermal loading. They showed that the best graphene platelets distribution is when they accumulate near the structure's upper and

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lower surface layer. They also showed that using this pattern enhances critical temperature. The core of the sandwich structure plays an important role in the strength, rigidity, and mechanical properties against the applied loads. The sandwich structure's core has low strength and high thickness. High thickness in the core of the sandwich structure causes high flexural stiffness and low density in this structure. Various studies have been performed on the behavior of sandwich structures with various types of cores, including foam core [28, 29], honeycomb core [30, 31], and balsa core [32]. Each of these cores has its unique capabilities. Therefore, a suitable core can be selected depending on whether the sandwich structure is used in the marine, automotive, aerospace, or civil industries. Smart materials such as elastomers, visco-elastic cores [33], and fluids have recently been used as cores in sandwich structures. Various researchers have investigated the vibrational behavior of sandwich structures with elastomer [34–37] and fluid [38–40] magnetorheological cores. When the magnetorheological core is exposed to the external magnetic field, its stiffness and strength are changed; in other words, the shear modulus of these materials changes and causes changes in the structure's features. Elastomers show a faster response in comparison with fluids in the magnetic field. Moreover, these materials do not have the disadvantages of magnetorheological fluids, such as sealing problems, particle precipitation, and environmental effects. Residual stress is inevitable during the production process of composite structures. These stresses may be caused by plastic deformation and thermal or structural changes (phase transformation) in the structure. Residual stress is sometimes unwantedly formed in structures because of welding [41], casting, and machining processes. Also, residual stresses are sometimes created in the structure using processes such as shot pinning [42–44] and autofrettage [45, 46] to improve the mechanical properties of that structure. Hashemi Kachapi [47] investigated the effects of surface/interface energy for tensile instability analysis on dimensionless natural frequency and nonlinear dynamic response of piezoelectric nanosensors under nonlinear electrostatic excitation and harmonic force were investigated using Gortin-Murdoch. The relationships were calculated based on Lagrange–Euler's theories and Hamilton's principle. He also used arc-length continuation and complex averaging methods to check various parameters and residual stress. The stress could be tensile or compressive and can cause beneficial or destructive consequences on the structures. Residual stress can also affect the stiffness and natural frequencies of the structures [48, 49]. Jiang et al. [50] studied vibration for piezoelectric nanobeams with internal residual stress. They showed that residual stress affects the frequency of the nanobeam and can change it up to 80 percent. Hence, residual stress in structures would affect the dynamic behavior, which should be considered when studying their vibrational behavior. Lin and Chen [51] and Karimi-Nobandegani et al. [52] used dynamic analysis/stability for a spinning pre-twisted structure with a visco-elastic core subjected to a periodic axial load.

This research investigates the effect of thermomechanical loading and residual stress on the dynamic response for a spinning visco-elastic structure with an MRE core and nanocomposite layers reinforced with GPLs using the Donnell-Moshtari theory. The face sheet layers for the sandwich cylindrical shell are considered metal matrix nanocomposites; furthermore, the GPLs are used with uniform FG distribution in the direction of thickness to reinforce these layers. The core layer is considered to be MRE based on the FSDT. The properties of MRE materials, GPLs, and metal matrix nanocomposites are assumed to depend on temperature. The nanocomposite properties are obtained based on the HT micromechanics model and ROM. The equations of motion for a spinning sandwich shell are driven by considering the rotary inertia and shear effect. The frequencies of the structure are obtained using the DQM. Moreover, the effect of parameters such as weight fraction and distribution of GPLs, MRE core, spinning speed, residual stresses, and thermomechanical loading on the dynamic response of a shell are studied.

2 Geometric models and basic formulations

2.1 Strain displacement

The spinning sandwich cylindrical shell is shown in Fig. 1. Also, the core is considered an MRE smart material. MRE core's mechanical properties differ based on whether the shear stress is the before or after yielding region. This analysis assumes that the MRE core is the before-yielding region and exhibits the behavior of linear visco-elastic materials [53]. Since the elastic modulus of the MRE core is very small compared to the elastic modulus of the inner and outer layers of the cylindrical sandwich shell, the normal stress in the core can be ignored.

All theories based on Kirchhoff Love assumptions, such as Novozhilov, Sanders, Flügge Lur'e Byrne, and Donnell's theories, are named the classical theories and are used to analyze elastic shells in equations.

The assumptions of Donnell's shell theory are stated as follows:

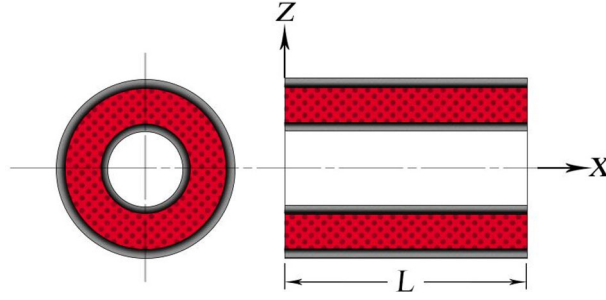


Fig. 1 A schematic view for a spinning sandwich structure with MRE smart core and nanocomposite skin layers reinforced by GPLs

- (a) The shell is considered as thin.
- (b) The strains are everywhere small.
- (c) The state of stress is approximately plane. Moreover, the transverse normal stress is small compared with other normal stress components and may be neglected.
- (d) Normals to the undeformed middle surface remain straight and also, normal to the deformed middle surface remains straight.
- (e) The displacements of an arbitrary point of a shell are small in comparison with its thickness.

In sandwich structures, the thickness of each face sheet layers is small in comparison with the thickness of core layer, thus based on above assumptions, for each face sheet layers, Donnell's shell theory is considered, while because the higher thickness for core layer with respect to face sheet layer, the first-order shear deformation shell theory is considered. Donnell's shell theory is an approximate theory specifically for cylinders that is proposed for a general intermediate surface. This theory is used for what can be called "pseudo-shallow shells." These theories can only be formulated for thin shells by not considering rotational inertia and shear deformation. According to the things mentioned in this work, the thickness of the inner and outer layers of the shell is very small compared to the length and radius of the shell. As a result, Donnell's theory can be used to describe the displacement field of the inner and outer layers. According to this theory, the slope at any point is very small, so in this theory, shear strains can be omitted, and complex equations can be simplified. Since the effects of rotational movement are considered for the shell in this article, and the authors choose Donnell's theory among the thin shell theories.

The only limitation or uncertainties in Donnell's shell theory is that it cannot be used for thick shells. According to this theory, the slope at any point is minimal. Therefore, shear strains γ_{xz} and $\gamma_{\theta z}$ can be omitted in this theory. Based on the assumptions above, the displacement field for a sandwich shell is expressed as follows [54]:

$$\begin{aligned}\bar{u}_i(x, \theta, z, t) &= u^i(x, \theta, t) - z_i \frac{\partial w(x, \theta, t)}{\partial x} \\ \bar{v}_i(x, \theta, z, t) &= v^i(x, \theta, t) - z_i \frac{\partial w(x, \theta, t)}{R_i \partial \theta} \quad (i = t, b) \\ \bar{w}_i(x, \theta, z, t) &= w(x, \theta, t)\end{aligned}\quad (1)$$

The displacement field for a spinning cylindrical sandwich shell core layer based on the first-order shear deformation shell theory (FSDST) is expressed as follows [54].

$$\begin{aligned}\bar{u}_c(x, \theta, z, t) &= u^c(x, \theta, t) + z_c \phi_x^c(x, \theta, t) \\ \bar{v}_c(x, \theta, z, t) &= v^c(x, \theta, t) + z_c \phi_\theta^c(x, \theta, t) \\ \bar{w}_c(x, \theta, z, t) &= w(x, \theta, t)\end{aligned}\quad (2)$$

The elastic modulus of the core is very small compared to the upper and lower layers of the structure, so the core's normal stresses can be ignored. According to the assumptions of the problem, there is no shear stress between the upper and lower layers and the core. It is also assumed that all layers have the same transverse displacement. Based on the proposed assumptions, as well as the displacement continuity at the interface of three layers of the rotating sandwich structure (between top face sheet and core ($u^t(z_t = -\frac{h_t}{2}) = u^c(z_c = +\frac{h_c}{2})$),

$v^t\left(z_t = -\frac{h_t}{2}\right) = v^c\left(z_c = +\frac{h_c}{2}\right)$ and between bottom face sheet and core ($u^b\left(z_b = +\frac{h_b}{2}\right) = u^c\left(z_c = -\frac{h_c}{2}\right)$, $v^b\left(z_b = +\frac{h_b}{2}\right) = v^c\left(z_c = -\frac{h_c}{2}\right)$), thus the following relations can be shown:

$$\begin{aligned} u^t\left(x, y, z_t = -\frac{h_t}{2}, t\right) &= u^c\left(x, y, z_c = +\frac{h_c}{2}, t\right) \\ v^t\left(x, y, z_t = -\frac{h_t}{2}, t\right) &= v^c\left(x, y, z_c = +\frac{h_c}{2}, t\right) \end{aligned} \quad (3a)$$

between top face sheet and core

$$\begin{aligned} u^c\left(x, y, z_c = -\frac{h_c}{2}, t\right) &= u^b\left(x, y, z_b = +\frac{h_b}{2}, t\right) \\ v^c\left(x, y, z_c = -\frac{h_c}{2}, t\right) &= v^b\left(x, y, z_b = +\frac{h_b}{2}, t\right) \end{aligned} \quad (3b)$$

between bottom face sheet and core.

Based on the continuity of displacements at the interface of layers by substituting Eqs. (1) and (2) into Eqs. (3a) and (3b), the following equations are established as:

$$\begin{aligned} u^t(x, \theta, t) - z_t \frac{\partial w(x, \theta, t)}{\partial x} &= u^c(x, \theta, t) + z_c \phi_x^c(x, \theta, t) \\ v^t(x, \theta, t) - z_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} &= v^c(x, \theta, t) + z_c \phi_\theta^c(x, \theta, t) \\ u^b(x, \theta, t) - z_b \frac{\partial w(x, \theta, t)}{\partial x} &= u^c(x, \theta, t) + z_c \phi_x^c(x, \theta, t) \\ v^b(x, \theta, t) - z_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} &= v^c(x, \theta, t) + z_c \phi_\theta^c(x, \theta, t) \end{aligned} \quad (3c)$$

Therefore, from the solution of Eq. (3), the core layer's displacement and rotation are obtained as follows.

$$\begin{aligned} u^c(x, \theta, t) &= \left(\frac{1}{2}\right) \left(u^t(x, \theta, t) + u^b(x, \theta, t)\right) + \left(\frac{h_t - h_b}{4}\right) \frac{\partial w(x, \theta, t)}{\partial x} \\ v^c(x, \theta, t) &= \left(\frac{1}{2}\right) \left(v^t(x, \theta, t) + v^b(x, \theta, t)\right) + \left(\frac{1}{4}\right) \left(h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} - h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta}\right) \\ \phi_x^c(x, \theta, t) &= \frac{u^t(x, \theta, t) - u^b(x, \theta, t)}{h_c} + \frac{h_t + h_b}{2h_c} \frac{\partial w(x, \theta, t)}{\partial x} \\ \phi_\theta^c(x, \theta, t) &= \frac{v^t(x, \theta, t) - v^b(x, \theta, t)}{h_c} + \left(\frac{1}{2h_c}\right) \left(h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} + h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta}\right) \end{aligned} \quad (4)$$

2.2 Magnetorheological elastomer core

Magnetorheological elastomer (MRE) is a composite structure consisting of magnetic particles and polymers in the role of the matrix, which changes the structure's mechanical properties due to the movements of these particles under the magnetic field. Magnetic particles play a significant role in improving the performance of MREs. However, it should be noted that the excessive increase in these particles can also be destructive and cause a decrease in the structure's tensile strength. According to the assumption for the core layer, we consider the behavior of the MRE as linear visco-elastic. The relationship between shear modulus and loss coefficient based on natural rubber (60% iron particles, 20% softener, and 20% matrix) strongly depends on the magnetic field. The magnetic field is applied to the sandwich structure in the z -direction. Thus, the shear stress components of the core are considered as follows.

$$\tau_{xz}^c = k_s G_c \gamma_{xz}^c, \quad \tau_{z\theta}^c = k_s G_c \gamma_{z\theta}^c \quad (5)$$

k_s is the shear factor, which value is considered for the cylindrical section $k_s = 5/6$. Also, γ_{xz}^c , $\gamma_{z\theta}^c$, and G_c represent shear strain and mixed shear modulus of magnetorheological core, respectively. The shear strain of the MRE core is expressed as the following equation:

$$\begin{aligned}\gamma_{xz}^c &= \frac{\partial \bar{w}_c(x, \theta, z, t)}{\partial x} + \frac{\partial \bar{u}_c(x, \theta, z, t)}{\partial z} \\ \gamma_{z\theta}^c &= \frac{1}{R_c} \left[\frac{\partial \bar{w}_c(x, \theta, z, t)}{\partial \theta} - \bar{v}_c(x, \theta, z, t) \right] + \frac{\partial \bar{v}_c(x, \theta, z, t)}{\partial z}\end{aligned}\quad (6)$$

The shear strain components of the MRE core in terms of displacements for a cylindrical sandwich shell are expressed by substituting Eqs. (2, 4) into Eq. (6).

$$\begin{aligned}\gamma_{xz}^c &= \left(\frac{1}{h_c} \right) (u^t(x, \theta, t) - u^b(x, \theta, t)) + \left(\frac{h_t + 2h_c + h_b}{2h_c} \right) \frac{\partial w(x, \theta, t)}{\partial x} \\ \gamma_{z\theta}^c &= \frac{1}{R_c} \left[\frac{\partial w(x, \theta, t)}{\partial \theta} - \left(\frac{1}{2} \right) (v^t(x, \theta, t) + v^b(x, \theta, t)) - \left(\frac{1}{4} \right) \left(h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} - h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} \right) \right] \\ &\quad + \left(\frac{1}{h_c} \right) (v^t(x, \theta, t) - v^b(x, \theta, t)) + \left(\frac{1}{2h_c} \right) \left(h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} + h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} \right)\end{aligned}\quad (7)$$

The complex shear modulus of the MRE core is expressed as follows.

$$G_c = G' (1 + i L_f) \quad (8)$$

At room temperature, the MRE core behaves like an elastomer, and with the increase in temperature, it changes behavior and turns into a soft viscoplastic material. As the temperature increases, the core matrix becomes more complicated due to the increasing role of chemical bonding. Therefore, considering that the cylindrical sandwich shell is under the temperature field, G' , and L_f represents the intensity of the storage modulus. The loss coefficients, respectively, whose values are based on the data obtained from the experiments by Chen et al. [55] at different temperatures and based on the magnetic field intensity, are presented. Chen et al. [55] by the experimental results presented as experimental diagrams. The equations in Table 1 are obtained by calculating the slope of the line of these graphs. The following figures are the results of experiments by Chen et al. [55] at different temperatures and based on the intensity of the magnetic field. With the help of mathematical software and curve fitting, the formulas for G' , and L_f of these data is calculated, and as a result, it is presented as a formula for using different modes. These graphs calculate the value of G' , and L_f in the said states. After placing these two values in Eq. (8), the complex shear modulus of the MRE core is obtained. On the other hands, based on experimental data by Chen et al. [55], the authors presented formulas for G' , and L_f using curve fitting using various temperatures that the accuracy of the formulas is shown in the following Figures and the obtained results in Table 1 are illustrated.

Based on Tables 1 and 3, the effective coefficients in the mechanical properties of the core and matrix of the procedures are presented. Based on these two tables and Eqs. (8) and (11), the properties of the core and the

Table 1 Values of storage shear modulus and loss factor at different temperatures

Temperature	Storage shear modulus and the loss factor
T@60 °C (333 K)	$G' = -6.9069(M_f)^6 + 23.382(M_f)^5 - 28.818(M_f)^4 + 14.956(M_f)^3 - 2.7325(M_f)^2 + 0.4028(M_f) + 1.0371$ $L_f = -0.4851(M_f)^6 + 1.6928(M_f)^5 - 2.2076(M_f)^4 + 1.3207(M_f)^3 - 0.3707(M_f)^2 + 0.0659(M_f) + 0.0657$
T@80 °C (353 K)	$G' = 3.5069(M_f)^6 - 23.84(M_f)^5 + 48.705(M_f)^4 - 41.523(M_f)^3 + 13.793(M_f)^2 - 0.0646(M_f) + 1.0625$ $L_f = 0.8939(M_f)^6 - 2.7233(M_f)^5 + 3.1604(M_f)^4 - 1.5882(M_f)^3 + 0.1326(M_f)^2 + 0.1638(M_f) + 0.1136$
T@100 °C (373 K)	$G' = -12.175(M_f)^6 + 36.135(M_f)^5 - 37.575(M_f)^4 + 15.01(M_f)^3 - 1.6053(M_f)^2 + 0.6413(M_f) + 1.0444$ $L_f = -1.7292(M_f)^6 + 5.4543(M_f)^5 - 6.6815(M_f)^4 + 4.1173(M_f)^3 - 1.4163(M_f)^2 + 0.2865(M_f) + 0.0852$
T@120 °C (393 K)	$G' = -12.431(M_f)^6 + 39.091(M_f)^5 - 44.7(M_f)^4 + 21.88(M_f)^3 - 4.24(M_f)^2 + 0.722(M_f) + 1.0399$ $L_f = 1.1597(M_f)^6 - 3.78(M_f)^5 + 4.589(M_f)^4 - 2.4889(M_f)^3 + 0.5179(M_f)^2 + 0.0156(M_f) + 0.0745$

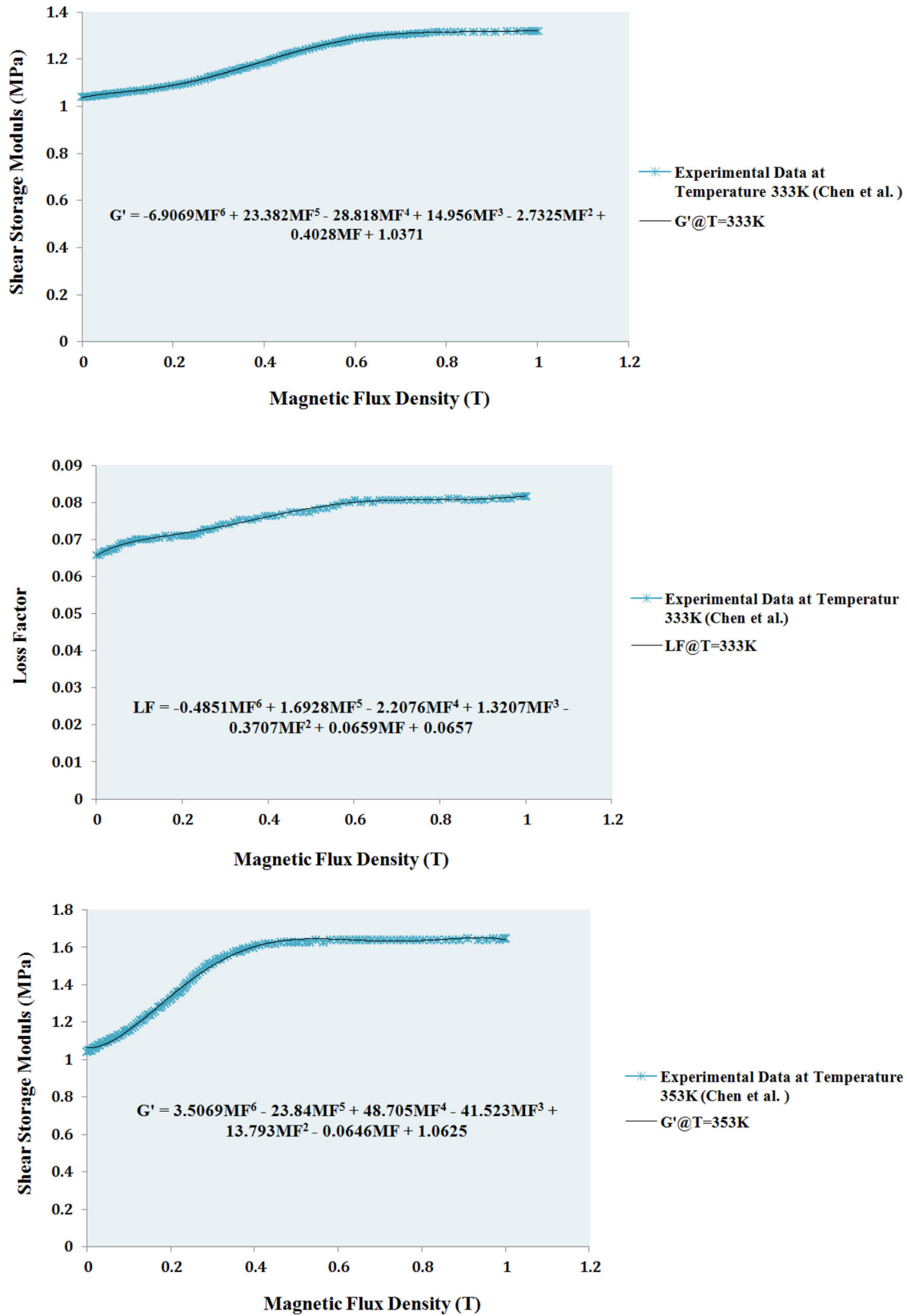


Fig. 2 Based on experimental data by Chen et al. [55], the presented formulas for G' , and L_f using curve fitting by authors for various temperatures

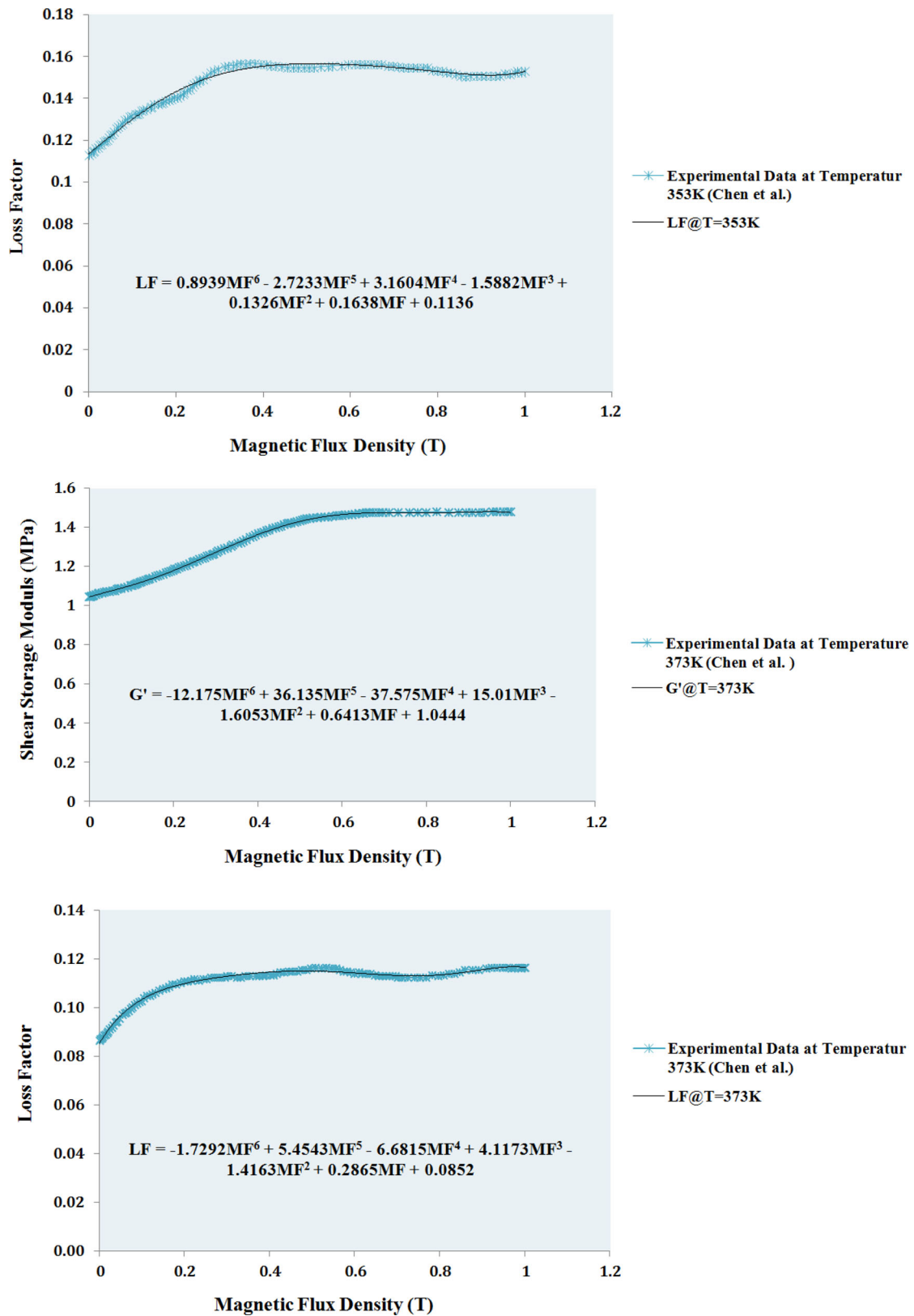


Fig. 2 continued

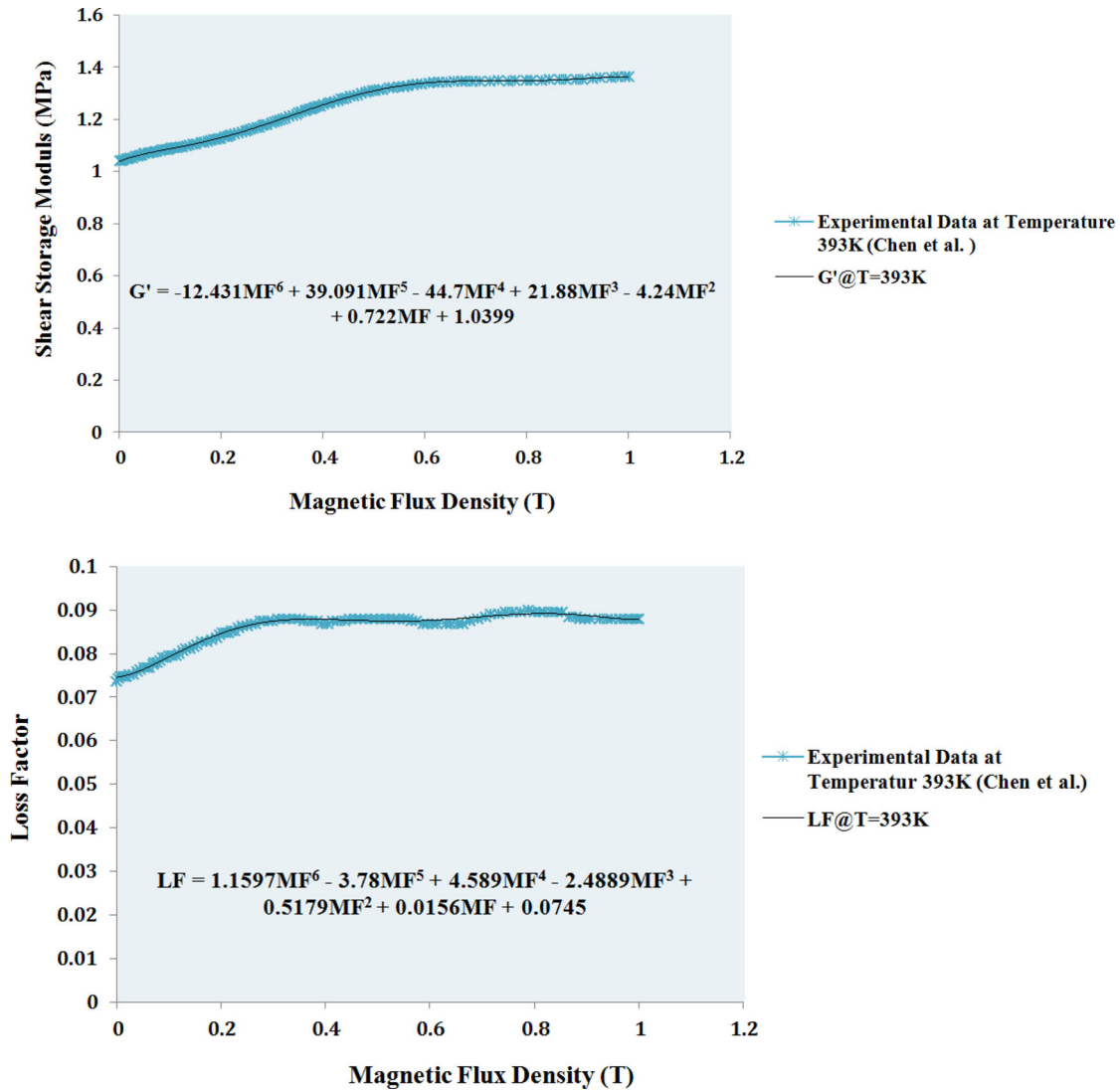


Fig. 2 continued

face sheet layers that are affected by temperature are obtained. Moreover, the effect of temperature on storage shear modulus and loss factor for core layer is plotted in Fig. 2. Also, the effect of temperature on Young's modulus of face sheet layers is illustrated in Fig. 3. According to Figs. 2 and 3, with increasing of temperature, the storage shear modulus and loss factor for core and Young's modulus decrease, because the stiffness of a sandwich cylindrical shell reduces, then the natural frequency decreases. The core considered for the structure is a smart magnetorheological core that can behave like an elastomer at room temperature and then change its behavior due to the increase in temperature and become a soft viscoplastic material. As the temperature increases, the core matrix leads to decrease the hardness due to the increasing role of chemical cross-linking. Therefore, considering that the cylindrical shell of the rotating sandwich is under the temperature field, it is very important to consider the effect of temperature on the mechanical properties of the core. In addition, the inner and outer layers of the rotating sandwich structure are made of metal matrix nanocomposite reinforced with graphene sheets. Since these structures are used in environments with a temperature gradient, therefore, to achieve a more accurate analysis of the properties of the material, they should be considered as a function of temperature. As a result, the mechanical properties of reed structures depend on temperature and undergo changes with temperature.

Table 1 shows the formulas obtained from the results of experiments by Chen et al. [55]. The experimental results obtained from the experiments of Chen et al. [55] are presented in the form of diagrams. Using

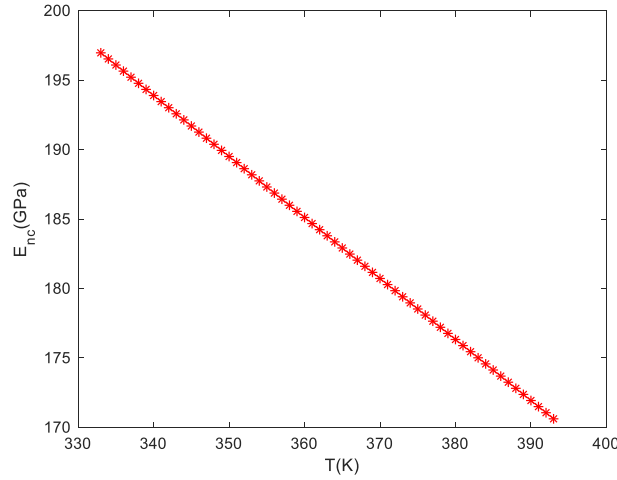


Fig. 3 Effect of temperature on the Young's modulus nanocomposite reinforced by GPL

mathematical programs and curve fitting by authors in Fig. 2, the storage shear modulus and the loss factor is calculated and presented in the form of formulas in Table 1.

2.3 Nanocomposite layers for a sandwich shell

Nanocomposite matrix is a metal base and one of the most common titanium alloys (Ti-6Al-4V) that is used in various industries due to its low density, specific strength, and excellent corrosion resistance. GPLs have increased to strengthen the structure. The Halpin–Tsai model estimates the mechanical properties of the cylindrical sandwich shell's inner and outer nanocomposite layers. Based on this model, the effective Young's modulus of the inner and outer layers of the cylindrical sandwich shell is expressed as follows.

$$E_{nc}^i = \left(\frac{3}{8} \left(\frac{1 + V_{GPL} e_L a_L}{1 - V_{GPL} e_L} \right) + \frac{5}{8} \left(\frac{1 + V_{GPL} a_B e_B}{1 - V_{GPL} e_B} \right) \right) E_M, \quad a_L = 2 \left(\frac{l_{GPL}}{t_{GPL}} \right), \quad a_B = 2 \left(\frac{b_{GPL}}{t_{GPL}} \right)$$

$$e_L = \left(\frac{\frac{E_{GPL}}{E_M} - 1}{\frac{E_{GPL}}{E_M} + a_L} \right), \quad e_B = \left(\frac{\frac{E_{GPL}}{E_M} - 1}{\frac{E_{GPL}}{E_M} + a_B} \right), \quad \Lambda_{GPL} = \frac{W_{GPL}}{W_{GPL} + \left(\frac{\rho_{GPL}}{\rho_M} \right) (1 - W_{GPL})} \quad (9)$$

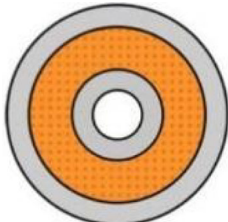
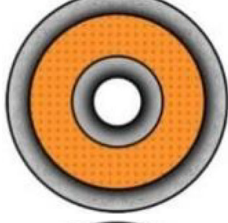
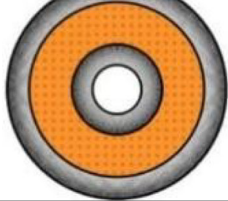
E_M , E_{GPL} , V_{GPL} , W_{GPL} , ρ_{GPL} , and ρ_M denote the elastic modulus of the matrix and GPL, the volume percentage, weight percentage, density of the GPL, and the density of the matrix, respectively. Also, the density values of ρ_i and Poisson's ratio ν of the cylindrical sandwich shell are expressed as follows based on the rule of mixture (ROM). In the Eq. (10), the law of mixing is a homotopy equation. Homotopy equations are equations in mathematics that change a geometric shape using equations continuously from one geometric shape to another. In other words, homotopy provides a way to classify topological spaces based on their structural similarities. By examining these transformations, mathematicians can understand how spaces are related and classify them into homotopy groups. These groups provide a powerful tool for distinguishing between different types of spaces. In Eq. 10, two nano-spaces (GPL) and macro (matrix), these equations are the way to convert these two spaces into one another [56].

$$\rho_{nc}^i = \rho_{GPL} V_{GPL} + \rho_m V_m, \quad \nu_{nc}^i = \nu_{GPL} V_{GPL} + \nu_m V_m, \quad \alpha_{nc}^i = \alpha_{GPL} V_{GPL} + \alpha_m V_m \quad (10)$$

In which, $V_m = (1 - V_{GPL})$, V_{GPL} , ν_{GPL} , ν_m , α_{GPL} , and α_m are the volume fractions of the matrix and GPL, Poisson's ratio of GPL and matrix, and the coefficient of thermal expansion of the GPL and the matrix, respectively.

The geometry of the GPLs is considered a rectangular solid section with a width b_{GPL} , length l_{GPL} , and thickness t_{GPL} . Different modes have been considered uniform and FG for the distribution of GPLs of the cylindrical sandwich shell. The schematic and relationships governing the distribution of GPLs of the spinning cylindrical sandwich shell for different modes are shown in Table 2.

Table 2 Governing relationships and distribution of GPLs in the inner and outer layers of the cylindrical sandwich shell

Type	Distribution of GPL	Domain	Schematic
U-U	$V_{\text{GPL}} = \Lambda_{\text{GPL}}$	$-\frac{h_i}{2} < z_i < +\frac{h_i}{2}$	
X-X	$V_{\text{GPL}} = \left(1 - \cos\left(\frac{\pi z_i}{h_i}\right)\right) \Lambda_{\text{GPL}}$		
O-O	$V_{\text{GPL}} = \cos\left(\frac{\pi z_i}{h_i}\right) \Lambda_{\text{GPL}}$		
V-Λ	$V_{\text{GPL}} = \begin{cases} \cos\left(\frac{\pi}{4} + \frac{\pi z_t}{2h_t}\right) \Lambda_{\text{GPL}} \\ \left(1 - \cos\left(\frac{\pi}{4} + \frac{\pi z_b}{2h_b}\right)\right) \Lambda_{\text{GPL}} \end{cases}$		
Λ-V	$V_{\text{GPL}} = \begin{cases} \left(1 - \cos\left(\frac{\pi}{4} + \frac{\pi z_t}{2h_t}\right)\right) \Lambda_{\text{GPL}} \\ \cos\left(\frac{\pi}{4} + \frac{\pi z_b}{2h_b}\right) \Lambda_{\text{GPL}} \end{cases}$		

The inner and outer layers of the rotating sandwich cylindrical shell are considered to be nanocomposite reinforced with graphene sheets. The nanocomposite matrix is a metal base and one of the most common titanium alloys (Ti 6Al 4V). Materials that are mainly a mixture of ceramic and metal are FG. The components of FG used in this research have temperature-dependent properties. Therefore, the properties of ceramics and metals used in the structure are expressed as a function of temperature. FGM materials are composite materials that are microscopically homogeneous, in which the mechanical properties vary smoothly and continuously from one surface to another. This is achieved by gradually changing the volume fraction of the constituent materials. This continuous change in composition results in graded properties of FGMs. The formulas used in this paper are based on the experimental data reported by Touloukian, which were presented as formulas by

Table 3 Temperature-dependent coefficients of material properties for metal matrix and GPL

Material properties		Γ_{-1}	Γ_0	Γ_{+1}	Γ_{+2}	Γ_{+3}
Metal Matrix	E	0	122.56×10^9	-4.586×10^{-4}	0	0
GPLs		0	1.01×10^{12}	-3.4×10^{-3}	0	0
Metal Matrix	ρ	0	4420	0	0	0
GPLs		0	1062.5	0	0	0
Metal Matrix	ν	0	0.2884	1.12×10^{-4}	0	0
GPLs		0	0.186	0	0	0
Metal Matrix	α	0	7.5788×10^{-6}	6.638×10^{-4}	-3.147×10^{-6}	0
GPLs		0	5×10^{-6}	0	0	0

Reddy and Chin [57]. Thus, the mechanical properties of the inner and outer layers of structure $\Gamma(T)$, which are obtained based on the experimental data [57], are expressed as the following equation:

$$\Gamma(T) = \Gamma_0 [\Gamma_{-1} T^{-1} + 1 + \Gamma_1 T^1 + \Gamma_2 T^2 + \Gamma_3 T^3] \quad (11)$$

The constant coefficients cubic proportions, the property of matter is constant. The material properties are expressed in such a way that the effects of higher temperatures on the material properties can be easily recognized. Each of the constant coefficients Γ_i ($i = -1, 0, 1, 2, 3$) has been experimentally obtained for different materials, as stated in Table 3. Also, the desired temperature is Kelvin for T . The relationship between stress and strain of the nanocomposite shell under the thermal field is expressed as follows.

In this article, the matrix used for metal face sheets is one of the most common titanium alloys (Ti 6Al 4V). Due to the fact that FG sandwich structure procedures are assumed, the related equations are considered as Eq. (11), and the constants of this equation are shown in Table 3. This table is based on Reddy and Chin [57] and the experimental data that is shown in Eq. (11) and Table 3. Also, the considered core models, described in Eq. (8) and Table 1, are obtained based on the experimental tests of Chen et al. [55]. After analyzing the experimental data using mathematics software, the best equation is obtained and presented based on the experimental results.

$$\begin{bmatrix} \sigma_x^i \\ \sigma_\theta^i \\ \tau_{x\theta}^i \end{bmatrix} = \begin{pmatrix} Q_{11}^i & Q_{12}^i & 0 \\ Q_{21}^i & Q_{22}^i & 0 \\ 0 & 0 & Q_{33}^i \end{pmatrix} \begin{bmatrix} \varepsilon_x^i - \alpha_{nc}^i \Delta T^i \\ \varepsilon_\theta^i - \alpha_{nc}^i \Delta T^i \\ \gamma_{x\theta}^i \end{bmatrix} \quad i = t, b \quad (12)$$

α^i and ΔT^i are thermal expansion coefficients and a function of temperature changes in the sandwich structure. It also Q_{mn}^i represents the matrix of elastic coefficients of the nanocomposite sandwich cylindrical shell, defined as the following equation:

$$Q_{11}^i = Q_{22}^i = \frac{E_{nc}^i}{1 - (\nu_{nc}^i)^2}, \quad Q_{12}^i = Q_{21}^i = \frac{E_{nc}^i \nu_{nc}^i}{1 - (\nu_{nc}^i)^2}, \quad Q_{33}^i = \frac{E_{nc}^i}{2(1 + \nu_{nc}^i)} \quad (13)$$

Also, the general strain–displacement equations for sandwich shells are expressed as follows.

$$\begin{aligned} \varepsilon_x^i &= \frac{\partial \bar{u}_i(x, \theta, z, t)}{\partial x} = \frac{\partial u^i(x, \theta, t)}{\partial x} - z_i \frac{\partial^2 w(x, \theta, t)}{\partial x^2} \\ \varepsilon_\theta^i &= \frac{\bar{w}_i(x, \theta, z, t)}{R_i} + \frac{1}{R_i} \frac{\partial \bar{v}_i(x, \theta, z, t)}{\partial \theta} = \frac{w(x, \theta, t)}{R_i} + \frac{1}{R_i} \left(\frac{\partial v^i(x, \theta, t)}{\partial \theta} - \frac{z_i}{R_i} \frac{\partial^2 w(x, \theta, t)}{\partial \theta^2} \right) \\ \gamma_{x\theta}^i &= \frac{1}{R_i} \frac{\partial \bar{u}_i(x, \theta, z, t)}{\partial \theta} + \frac{\partial \bar{v}_i(x, \theta, z, t)}{\partial x} = \frac{1}{R_i} \frac{\partial u^i(x, \theta, t)}{\partial \theta} + \frac{\partial v^i(x, \theta, t)}{\partial x} - 2 \frac{z_i}{R_i} \frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \end{aligned} \quad (14)$$

3 The equations of motion for sandwich shell

The energy method has been used among the various techniques that exist to obtain the equations of motion for the spinning nanocomposite sandwich cylindrical shell that is defined as

$$\Pi = [K - U + W] \quad (15)$$

K , U , and W are kinetic energy, strain energy, and work caused by external forces acting on the rotating nanocomposite sandwich cylindrical shell.

3.1 Strain energy

This article investigates the vibrations of a rotating nanocomposite sandwich cylindrical shell with an MRE core. According to assumptions, this sandwich shell is not under electric fields. Also, no piezoelectric layer is assumed in this structure. The authors have considered the structure's core MRE in the present work. One of the advantages of this material is its magnetic properties. For this reason, the magnetic field is assumed only inside the core, according to Table 1. Due to the mentioned reasons, the authors did not consider piezoelectric energy and the geometric potential of strain energy. Thus, the total strain energy includes the energy caused by stretching, bending, and energy stored in face sheet layers and the core U_s , as well as the energy caused by the residual stresses in the face sheet layers U_r .

$$U = U_s + U_r \quad (16)$$

The strain energy of the face sheet and core layers for a spinning nanocomposite cylindrical sandwich shell is expressed as follows.

$$U_s = \frac{1}{2} \int_0^{2\pi} \int_0^L \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} \left(\sigma_x^i \varepsilon_x^i + \sigma_\theta^i \varepsilon_\theta^i + \tau_{x\theta}^i \gamma_{x\theta}^i \right) R_i dz dx d\theta + \frac{1}{2} \int_0^{2\pi} \int_0^L \int_{-\frac{h_c}{2}}^{+\frac{h_c}{2}} \left(\tau_{xz}^c \gamma_{xz}^c + \tau_{z\theta}^c \gamma_{z\theta}^c \right) R_c dz dx d\theta \quad (17)$$

$\sigma_x^i, \sigma_\theta^i$, and $\tau_{x\theta}^i$ represent the stress terms, and $\varepsilon_x^i, \varepsilon_\theta^i$, and $\gamma_{x\theta}^i$ represent the strain terms in face sheet layers for the spinning nanocomposite sandwich cylindrical shell. The strain energy of the spinning nanocomposite sandwich cylindrical shell is obtained as follows.

$$\begin{aligned} U_s = & \frac{1}{2} \int_0^{2\pi} \int_0^L \left(N_x^i \frac{\partial u^i(x, \theta, t)}{\partial x} - M_x^i \frac{\partial^2 w(x, \theta, t)}{\partial x^2} \right. \\ & + N_\theta^i \frac{1}{R_i} \left(w(x, \theta, t) + \frac{\partial v^i(x, \theta, t)}{\partial \theta} \right) - M_\theta^i \frac{1}{R_i^2} \frac{\partial^2 w(x, \theta, t)}{\partial \theta^2} \\ & - 2M_{x\theta}^i \frac{1}{R_i} \frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} + N_{x\theta}^i \left(\frac{1}{R_i} \frac{\partial u^i(x, \theta, t)}{\partial \theta} + \frac{\partial v^i(x, \theta, t)}{\partial x} \right) \Big) R_i dx d\theta \\ & + \frac{1}{2} \int_0^{2\pi} \int_0^L \left(Q_{xz}^c \left(\frac{1}{h_c} \right) (u^t(x, \theta, t) - u^b(x, \theta, t)) \right. \\ & + \left(\frac{h_t + 2h_c + h_b}{2h_c} \right) \frac{\partial w(x, \theta, t)}{\partial x} + Q_{z\theta}^c \left(\frac{1}{R_c} \left[\frac{\partial w(x, \theta, t)}{\partial \theta} - \left(\frac{1}{2} \right) (v^t(x, \theta, t) + v^b(x, \theta, t)) \right] \right. \\ & - \left(\frac{1}{4} \right) \left(h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} - h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} \right) \Big) + \left(\frac{1}{h_c} \right) (v^t(x, \theta, t) - v^b(x, \theta, t)) \\ & \left. + \left(\frac{1}{2h_c} \right) \left(h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} + h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} \right) \right) R_c dx d\theta \end{aligned} \quad (18)$$

The force and moment components for the inner and outer layers and the core of the spinning nanocomposite shell are expressed as follows.

$$\begin{aligned} \begin{pmatrix} N_x^i \\ M_x^i \\ P_x^i \end{pmatrix} &= \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} \begin{pmatrix} \sigma_x^i \\ \sigma_x^i z_i \\ \sigma_x^i z_i^2 \end{pmatrix} dz, \quad \begin{pmatrix} N_\theta^i \\ M_\theta^i \\ P_\theta^i \end{pmatrix} = \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} \begin{pmatrix} \sigma_\theta^i \\ \sigma_\theta^i z_i \\ \sigma_\theta^i z_i^2 \end{pmatrix} dz, \\ \begin{pmatrix} N_{x\theta}^i \\ M_{x\theta}^i \\ P_{x\theta}^i \end{pmatrix} &= \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} \begin{pmatrix} \tau_{x\theta}^i \\ \tau_{x\theta}^i z_i \\ \tau_{x\theta}^i z_i^2 \end{pmatrix} dz, \quad d\zeta \begin{pmatrix} Q_{xz}^c \\ Q_{z\theta}^c \end{pmatrix} = \int_{-\frac{h_c}{2}}^{+\frac{h_c}{2}} \begin{pmatrix} \tau_{xz}^c \\ \tau_{z\theta}^c \end{pmatrix} dz \end{aligned}$$

$$\begin{aligned}
 N_{Tx}^i &= \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} (Q_{11}^i + Q_{12}^i) \alpha_{nc}^i \Delta T^i dz, \quad M_{Tx}^i = \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} (Q_{11}^i + Q_{12}^i) z_i \alpha_{nc}^i \Delta T^i dz, \\
 N_{T\theta}^i &= \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} (Q_{22}^i + Q_{21}^i) \alpha_{nc}^i \Delta T^i dz \\
 M_{T\theta}^i &= \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} (Q_{22}^i + Q_{21}^i) z_i \alpha_{nc}^i \Delta T^i dz, \quad \Delta T^i = T^i - T_0
 \end{aligned} \tag{19}$$

The strain energy caused by the residual stresses in the face sheet layers of the spinning nanocomposite sandwich cylindrical shell can appear as a deviation in the structure's overall structure. In this research, the residual stresses' effect is considered a symmetrical deviation in the z -direction, and the torsional deviation of the shell is also neglected.

$$\begin{aligned}
 U_r &= \frac{1}{2} \sigma_r \int_0^{2\pi} \int_0^L \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} \left[\left(\frac{\partial w(x, \theta, t)}{\partial x} \right)^2 + a_3 \left(\frac{\partial v^i(x, \theta, t)}{\partial x} \right)^2 \right] R_i dx d\theta dz \quad i = t, b, c \\
 U_r &= \frac{1}{2} \sigma_r \int_0^{2\pi} \int_0^L \left[(R_b h_b + R_c h_c + R_t h_t) \left(\frac{\partial w(x, \theta, t)}{\partial x} \right)^2 + a_3 (R_b h_b) \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right)^2 + a_3 (R_t h_t) \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right)^2 \right. \\
 &\quad \left. + a_3 (R_c h_c) \left[\left(\frac{1}{4} \right) \left(\left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right)^2 + \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right)^2 + 2 \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right) \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right) \right) \right] \right. \\
 &\quad \left. + \left(\frac{1}{16} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right)^2 + \left(\frac{1}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right) \right) \right. \\
 &\quad \left. + \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right) \right] dx d\theta
 \end{aligned} \tag{20}$$

σ_r is the residual stress in the spinning nanocomposite cylindrical sandwich shell. Substituting Eqs. (18, 20) into Eq. (16), the total strain energy of the spinning nanocomposite shell with MRE core is obtained as follows:

$$\begin{aligned}
 U &= \frac{1}{2} \int_0^{2\pi} \int_0^L \left(N_x^i \frac{\partial u^i(x, \theta, t)}{\partial x} - M_x^i \frac{\partial^2 w(x, \theta, t)}{\partial x^2} + N_{\theta}^i \frac{1}{R_i} \left(w(x, \theta, t) + \frac{\partial v^i(x, \theta, t)}{\partial \theta} \right) \right. \\
 &\quad \left. - M_{\theta}^i \frac{1}{R_i^2} \frac{\partial^2 w(x, \theta, t)}{\partial \theta^2} - 2M_{x\theta}^i \frac{1}{R_i} \frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} + N_{x\theta}^i \left(\frac{1}{R_i} \frac{\partial u^i(x, \theta, t)}{\partial \theta} + \frac{\partial v^i(x, \theta, t)}{\partial x} \right) \right) R_i dx d\theta \\
 &\quad + \frac{1}{2} \int_0^{2\pi} \int_0^L \left(Q_{xz}^c \left(\left(\frac{1}{h_c} \right) (u^t(x, \theta, t) - u^b(x, \theta, t)) + \left(\frac{h_t + 2h_c + h_b}{2h_c} \right) \frac{\partial w(x, \theta, t)}{\partial x} \right) \right. \\
 &\quad \left. + Q_{z\theta}^c \left(\frac{1}{R_c} \left[\frac{\partial w(x, \theta, t)}{\partial \theta} - \left(\frac{1}{2} \right) (v^t(x, \theta, t) + v^b(x, \theta, t)) - \left(\frac{1}{4} \right) \left(h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} - h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} \right) \right] \right) \right. \\
 &\quad \left. + \left(\frac{1}{h_c} \right) (v^t(x, \theta, t) - v^b(x, \theta, t)) + \left(\frac{1}{2h_c} \right) \left(h_b \frac{\partial w(x, \theta, t)}{R_b \partial \theta} + h_t \frac{\partial w(x, \theta, t)}{R_t \partial \theta} \right) \right) R_c dx d\theta \\
 &\quad + \frac{1}{2} \sigma_r \int_0^{2\pi} \int_0^L \left((R_b h_b + R_c h_c + R_t h_t) \left(\frac{\partial w(x, \theta, t)}{\partial x} \right)^2 + a_3 (R_b h_b) \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right)^2 + a_3 (R_t h_t) \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right)^2 \right. \\
 &\quad \left. + a_3 (R_c h_c) \left[\left(\frac{1}{4} \right) \left(\left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right)^2 + \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right)^2 + 2 \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right) \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right) \right) \right] \right) dx d\theta
 \end{aligned}$$

$$\begin{aligned}
& + \left(\frac{1}{16} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right)^2 + \left(\frac{1}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right) \right. \\
& \left. + \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right) \right) \Big] dx d\theta \quad i = t, b
\end{aligned} \tag{21}$$

$N_x^i, N_\theta^i, M_x^i, M_\theta^i, N_{x\theta}^i, Q_{xz}^c$ and $Q_{z\theta}^c$ are stated in Appendix A1.

3.2 Kinetic energy

The nanocomposite sandwich cylindrical shell analyzed in this research is spinning, so the kinetic energy due to spinning inertia and gyroscopic effects are expressed as follows [58].

$$T = \frac{1}{2} \int_0^{2\pi} \int_0^L \int_{-h_i/2}^{+h_i/2} \rho_{nc}^i \left(\left(\frac{\partial \bar{u}_i}{\partial t} \right)^2 + \left(\frac{\partial \bar{v}_i}{\partial t} \right)^2 + \left(\frac{\partial \bar{w}_i}{\partial t} \right)^2 + 2\Omega \left[\bar{v}_i \frac{\partial \bar{w}_i}{\partial t} - \bar{w}_i \frac{\partial \bar{v}_i}{\partial t} \right] + \Omega^2 [\bar{v}_i^2 + \bar{w}_i^2] \right) r d\theta dx dz \quad i = b, t, c \tag{22}$$

Substituting Eqs. (1, 2, 4) into Eq. (22), the kinetic energy of the spinning nanocomposite sandwich cylindrical shell is obtained as follows:

$$T = T_i + T_c$$

$$\begin{aligned}
T = & \frac{1}{2} \int_0^{2\pi} \int_0^L \left[I_0^i \left(\left(\frac{\partial u^i(x, \theta, t)}{\partial t} \right)^2 + \left(\frac{\partial v^i(x, \theta, t)}{\partial t} \right)^2 + 2\Omega \left(\left(\frac{\partial w(x, \theta, t)}{\partial t} \right) v^i(x, \theta, t) - \left(\frac{\partial v^i(x, \theta, t)}{\partial t} \right) w(x, \theta, t) \right) \right. \right. \\
& + \Omega^2 (v^i(x, \theta, t)^2 + w(x, \theta, t)^2) + \left. \left(\frac{\partial w(x, \theta, t)}{\partial t} \right) \right) \\
& + 2I_1^i \left(- \left(\frac{\partial u^i(x, \theta, t)}{\partial t} \right) \left(\frac{\partial w(x, \theta, t)}{\partial x \partial t} \right) - \frac{1}{R_i} \left(\frac{\partial v^i(x, \theta, t)}{\partial t} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial \theta \partial t} \right) - \frac{\Omega^2}{R_i} \left(v^i(x, \theta, t) \left(\frac{\partial w(x, \theta, t)}{\partial \theta} \right) \right) \right. \\
& + \left. \left. \frac{\Omega}{R_i} \left(w(x, \theta, t) \left(\frac{\partial^2 w(x, \theta, t)}{\partial \theta \partial t} \right) - \left(\frac{\partial w(x, \theta, t)}{\partial t} \right) \left(\frac{\partial w(x, \theta, t)}{\partial \theta} \right) \right) \right) \right. \\
& + \left. I_2^i \left(\left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right)^2 + \frac{1}{R_i^2} \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right)^2 + \frac{\Omega^2}{R_i^2} \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right)^2 \right) \right] R_i d\theta dx \\
& + \left[I_0^c \left(\frac{1}{4} \left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right)^2 + \frac{1}{4} \left(\frac{\partial}{\partial t} u^b(x, \theta, t) \right)^2 + \frac{1}{2} \left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right) \left(\frac{\partial}{\partial t} u^b(x, \theta, t) \right) \right. \right. \\
& + \frac{1}{4} \left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right)^2 + \frac{1}{2} \left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right) \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right) + \left. \left(\frac{\partial}{\partial t} w(x, \theta, t) \right)^2 \right. \\
& + \frac{1}{16} (h_t - h_b)^2 \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right)^2 + \frac{1}{4} \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right)^2 + \frac{1}{16} \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right)^2 \\
& + \frac{1}{4} \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right) \left(\left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right) + \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right) \right) \\
& + \Omega \left(\frac{\partial}{\partial t} w(x, \theta, t) \right) \left(v^t(x, \theta, t) + v^b(x, \theta, t) \right) + \frac{1}{4} (h_t - h_b) \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right) \left(\left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right) + \left(\frac{\partial}{\partial t} u^b(x, \theta, t) \right) \right) \\
& + \frac{1}{16} \Omega^2 \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right)^2 + \Omega^2 [w(x, \theta, t)]^2 - \Omega w(x, \theta, t) \left(\left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right) + \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right) \right) \\
& + \frac{1}{2} \Omega \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\left(\frac{\partial}{\partial t} w(x, \theta, t) \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right) - \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right) w(x, \theta, t) \right) + \frac{1}{2} \Omega^2 v^t(x, \theta, t) v^b(x, \theta, t) \\
& + \frac{1}{4} \Omega^2 \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right) \left(v^t(x, \theta, t) + v^b(x, \theta, t) \right) + \frac{1}{4} \Omega^2 \left((v^t(x, \theta, t))^2 + (v^b(x, \theta, t))^2 \right) \\
& + I_1^c \left(\frac{1}{4h_c} \left(\frac{h_t^2}{R_t^2} - \frac{h_b^2}{R_b^2} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial \theta \partial t} \right)^2 + \frac{h_t}{h_c} \left(\frac{\partial u^t(x, \theta, t)}{\partial t} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial t} \right) + \frac{h_b}{h_c} \left(\frac{\partial u^b(x, \theta, t)}{\partial t} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial t} \right) \right)
\end{aligned}$$

$$\begin{aligned}
 & + \frac{1}{4} \frac{h_t^2}{h_c} \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial t} \right)^2 + \frac{1}{h_c} \left(\left(\frac{\partial u^t(x, \theta, t)}{\partial t} \right)^2 - \left(\frac{\partial u^b(x, \theta, t)}{\partial t} \right)^2 + \left(\frac{\partial v^t(x, \theta, t)}{\partial t} \right)^2 - \left(\frac{\partial v^b(x, \theta, t)}{\partial t} \right)^2 \right) \\
 & - \frac{1}{4} \frac{h_b^2}{h_c} \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial t} \right)^2 + \frac{h_b}{h_c R_b} \left(\frac{\partial v^b(x, \theta, t)}{\partial t} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial \theta \partial t} \right) + \frac{h_t}{h_c R_t} \left(\frac{\partial v^t(x, \theta, t)}{\partial t} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial \theta \partial t} \right) \\
 & + \frac{\Omega}{h_c} \left(\frac{h_t}{R_t} + \frac{h_b}{R_b} \right) \left(\left(\frac{\partial w(x, \theta, t)}{\partial t} \right) \left(\frac{\partial w(x, \theta, t)}{\partial \theta} \right) - w(x, \theta, t) \left(\frac{\partial^2 w(x, \theta, t)}{\partial \theta \partial t} \right) \right) + \frac{\Omega^2 h_t}{h_c R_t} \left(v^t(x, \theta, t) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right) \right) \\
 & + \frac{2\Omega}{h_c} \left(\left(\frac{\partial}{\partial t} w(x, \theta, t) \right) \left(v^t(x, \theta, t) - v^b(x, \theta, t) \right) + w(x, \theta, t) \left(\left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right) - \left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right) \right) \right) \\
 & + \frac{\Omega^2 h_b}{h_c R_b} \left(v^b(x, \theta, t) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right) \right) + \frac{\Omega^2}{4h_c} \left(\frac{h_t^2}{R_t^2} - \frac{h_b^2}{R_b^2} \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right)^2 + \frac{\Omega^2}{h_c} \left(v^t(x, \theta, t)^2 - v^b(x, \theta, t)^2 \right) \\
 & + I_2^c \left(\frac{1}{h_c^2} \left(\left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right)^2 + \left(\frac{\partial}{\partial t} u^b(x, \theta, t) \right)^2 + \left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right)^2 + \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right)^2 \right) + \frac{1}{2} \frac{h_b h_t}{h_c^2 R_b R_t} \left(\left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right)^2 \right) \right. \\
 & + \frac{1}{4} \frac{h_b^2}{h_c^2 R_b^2} \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right)^2 - \frac{2}{h_c^2} \left(\left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right) \left(\frac{\partial}{\partial t} u^b(x, \theta, t) \right) + \left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right) \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right) \right) \\
 & + \frac{1}{4} \frac{h_t^2}{h_c^2 R_t^2} \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right)^2 + \frac{h_t}{h_c^2} \left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right) \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right) \\
 & + \frac{1}{4} \left(\frac{h_t^2}{h_c^2} + \frac{h_b^2}{h_c^2} \right) \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right)^2 + \frac{1}{2} \frac{h_t h_b}{h_c^2} \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right)^2 + \frac{\Omega^2}{4h_c^2} \left(\frac{h_b^2}{R_b^2} + \frac{h_t^2}{R_t^2} \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right)^2 \\
 & + \frac{\Omega^2}{h_c^2} \left(v^t(x, \theta, t)^2 + v^b(x, \theta, t)^2 \right) \frac{h_b}{h_c^2} \left(\frac{\partial}{\partial t} u^t(x, \theta, t) \right) \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right) - \left(\frac{h_t}{h_c^2} + \frac{h_b}{h_c^2} \right) \left(\frac{\partial}{\partial t} u^b(x, \theta, t) \right) \left(\frac{\partial^2}{\partial x \partial t} w(x, \theta, t) \right) \\
 & + \left(\frac{1}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \left(\frac{\partial^2}{\partial \theta \partial t} w(x, \theta, t) \right) \left(\left(\frac{\partial}{\partial t} v^t(x, \theta, t) \right) - \left(\frac{\partial}{\partial t} v^b(x, \theta, t) \right) \right) + \frac{1}{2} \frac{\Omega^2 h_b h_t}{h_c^2 R_b R_t} \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right)^2 \\
 & + \left(\frac{\Omega^2}{h_c^2} \right) \left(\frac{h_t}{R_t} + \frac{h_b}{R_b} \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right) \left(v^t(x, \theta, t) - v^b(x, \theta, t) \right) - \frac{2\Omega^2}{h_c^2} \left(v^t(x, \theta, t) v^b(x, \theta, t) \right) \left. \right] r dx d\theta \quad (23)
 \end{aligned}$$

I_j^i ($i = t, b, j = 0, 1, 2$) and I_j^c ($j = 0, 1, 2$) are the moments of inertia for the cylindrical sandwich shell's inner, outer, and core layers, which are defined as follows.

$$\begin{pmatrix} I_0^i \\ I_1^i \\ I_2^i \end{pmatrix} = \int_{-h_i/2}^{+h_i/2} \rho_{nc}^i \begin{pmatrix} 1 \\ z \\ z^2 \end{pmatrix} dz, \quad \begin{pmatrix} I_0^c \\ I_1^c \\ I_2^c \end{pmatrix} = \int_{-h_c/2}^{+h_c/2} \rho_c \begin{pmatrix} 1 \\ z \\ z^2 \end{pmatrix} dz, \quad J = 2I \quad (24)$$

3.3 Thermo-Mechanical external loading

As a result of the input energy loss caused by the external force, the vibrations of the rotating structure can be controlled. Considering that the analyzed sandwich structure has an MRE core and face sheet layers are also reinforced by GPLs, it is possible various thicknesses of core or the surface layers, as well as the distribution of GPL reinforcements to an optimal model, were obtained to minimize vibrations in the structure. The work due to axial loading is expressed as follows.

$$W_F = \frac{1}{2} \int_0^L \int_0^{2\pi} \int_{-\frac{h_i}{2}}^{+\frac{h_i}{2}} q_a \left[\left(\frac{\partial w(x, \theta, t)}{\partial x} \right)^2 + a_1 \left(\frac{\partial v^i(x, \theta, t)}{\partial x} \right)^2 \right] R dx d\theta \quad i = t, c, b \quad (25)$$

The mechanical properties of the sandwich structure depend on the temperature and change with the temperature change. The work done due to temperature changes is expressed as follows.

$$W_T = \frac{1}{2} \int_0^L \int_0^{2\pi} \left[N_{Tx}^i \left(\frac{\partial w(x, \theta, t)}{\partial x} \right)^2 + N_{T\theta}^i \left(\frac{\partial v^i(x, \theta, t)}{\partial x} \right)^2 \right] R_i dx d\theta,$$

$$N_{T_x}^c = N_{T_\theta}^c = \int_{-\frac{h_c}{2}}^{+\frac{h_c}{2}} 2G_c(1 + \nu_c)\alpha^c \Delta T^c dz, \quad i = t, (c), b \quad (26)$$

Substituting Eq. (4) into Eqs. (25, 26), the total external work of the spinning nanocomposite sandwich cylindrical shell, including the work caused by mechanical and thermal loading, is expressed as follows:

$$W = W_F + W_T$$

$$\begin{aligned} W = \frac{1}{2} \int_0^{2\pi} \int_0^L & \left[\left[q_f(h_b + h_c + h_t) + N_{T_x}^t + N_{T_x}^c + N_{T_x}^b \right] \left(\frac{\partial w(x, \theta, t)}{\partial x} \right)^2 + \left[a_1 q_f h_b + a_2 N_{T_\theta}^b \right] \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right)^2 \right. \\ & + \left[a_1 q_f h_t + a_2 N_{T_\theta}^t \right] \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right)^2 \\ & + \left[a_1 q_f h_c + a_2 N_{T_\theta}^c \right] \left[\left(\frac{1}{4} \right) \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right)^2 + \left(\frac{1}{4} \right) \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right)^2 \right. \\ & + \left(\frac{1}{2} \right) \left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right) \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right) \\ & + \left(\frac{1}{16} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right)^2 \left(\frac{1}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left[\left(\frac{\partial v^t(x, \theta, t)}{\partial x} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right) \right. \\ & \left. \left. + \left(\frac{\partial v^b(x, \theta, t)}{\partial x} \right) \left(\frac{\partial^2 w(x, \theta, t)}{\partial x \partial \theta} \right) \right] \right] R dx d\theta \quad (27) \end{aligned}$$

3.4 Hamilton's principle

Hamilton's principle states that the change with time of the total energy of the rotating sandwich cylindrical shell is zero. According to Hamilton's principle, inertial and elastic forces are not considered, and kinetic and potential energy are used instead. By using Hamilton's principle, the overall order of the governing equations of a system is reduced by one order; thus, the calculations become simpler. The equations obtained from this process are the governing equations of the system. This solution method is used to solve equations and obtain governing equations in all engineering sciences and mathematics [59]. Hamilton's principle states that the changes concerning the time of the total energy of the spinning sandwich cylindrical shell (Π) are equal to zero.

$$\int_0^T \delta [(T) - (U) + (W)] dt = 0 \quad (28)$$

K , U , and W express the kinetic, potential, and external work caused by the axial force acting on the spinning sandwich cylindrical shell, respectively. Substituting the Eqs. (21, 23, 27) into the equation of Hamilton's principle, using an integral part by part and arranging the relationships, the equation of motion for the spinning nanocomposite sandwich cylindrical shell with MRE core is obtained as follows.

$$\begin{aligned} \delta u^t : & -R_c \left(I_2^c \left(\frac{h_t + h_b}{h_c^2} \right) + I_0^c \left(\frac{h_t - h_b}{4} \right) \right) \left(\frac{\partial^3 w}{\partial x \partial t^2} \right) - 2 \left(\frac{B_{12}^t}{R_t} + 2 \frac{B_{33}^t}{R_t} \right) \left(\frac{\partial^3 w}{\partial x \partial \theta^2} \right) \\ & - 2R_t B_{11}^t \left(\frac{\partial^3 w}{\partial x^3} \right) - \left(2R_t I_0^t + \frac{1}{2} R_c I_0^c + \frac{2R_c I_2^c}{h_c^2} \right) \left(\frac{\partial^2 u^t}{\partial t^2} \right) \\ & + 2R_t A_{11}^t \left(\frac{\partial^2 u^t}{\partial x^2} \right) + 2 \frac{A_{33}^t}{R_t} \left(\frac{\partial^2 u^t}{\partial \theta^2} \right) \\ & + R_c \left(2 \frac{I_2^c}{h_c^2} - \frac{I_0^c}{2} \right) \left(\frac{\partial^2 u^b}{\partial t^2} \right) + 2(A_{12}^t + A_{33}^t) \left(\frac{\partial^2 v^t}{\partial x \partial \theta} \right) \\ & + 2 \left(A_{12}^t - \left(\frac{1}{h_c} \right) \alpha_1 R_c A_{11}^c \right) \left(\frac{\partial w}{\partial x} \right) + 2R_c A_{11}^c \left(\frac{1}{h_c^2} \right) (u^b - u^t) \end{aligned}$$

$$\begin{aligned}
 \delta u^b : & R_c \left(\left(\frac{h_t + h_b}{h_c^2} \right) I_2^c - \left(\frac{h_t - h_b}{4} \right) I_0^c \right) \left(\frac{\partial^3 w}{\partial x \partial t^2} \right) - 2 \left(\frac{B_{12}^b}{R_b} + 2 \frac{B_{33}^b}{R_b} \right) \left(\frac{\partial^3 w}{\partial x \partial \theta^2} \right) \\
 & - 2 R_b B_{11}^b \left(\frac{\partial^3 w}{\partial x^3} \right) + R_c \left(2 \frac{I_2^c}{h_c^2} - \frac{I_0^c}{2} \right) \left(\frac{\partial^2 u^t}{\partial t^2} \right) - \left(2 R_b I_0^b + \frac{1}{2} R_c I_0^c + \frac{2 R_c I_2^c}{h_c^2} \right) \left(\frac{\partial^2 u^b}{\partial t^2} \right) \\
 & + 2 R_b A_{11}^b \left(\frac{\partial^2 u^b}{\partial x^2} \right) + 2 \frac{A_{33}^b}{R_b} \left(\frac{\partial^2 u^b}{\partial \theta^2} \right) + 2 \left(A_{12}^b + A_{33}^b \right) \left(\frac{\partial^2 v^b}{\partial x \partial \theta} \right) \\
 & + 2 \left(A_{12}^b - \left(\frac{1}{h_c} \right) \alpha_1 R_c A_{11}^c \right) \left(\frac{\partial w}{\partial x} \right) + 2 R_c A_{11}^c \left(\frac{1}{h_c^2} \right) (u^t - u^b) \\
 \delta v^t : & - R_c \left(\frac{I_0^c}{4} \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) + \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) \left(\frac{\partial^3 w}{\partial \theta \partial t^2} \right) - 2 \frac{B_{22}^t}{R_t^2} \left(\frac{\partial^3 w}{\partial \theta^3} \right) \\
 & - \left(\frac{1}{4} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left[\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right] + 2 (B_{12}^t + 2 B_{33}^t) \right) \left(\frac{\partial^3 w}{\partial x^2 \partial \theta} \right) \\
 & + 2 (A_{12}^t + A_{33}^t) \left(\frac{\partial^2 u^t}{\partial x \partial \theta} \right) - \left(2 R_t I_0^t + \frac{1}{2} R_c I_0^c + \frac{2 R_c I_2^c}{h_c^2} \right) \left(\frac{\partial^2 v^t}{\partial t^2} \right) \\
 & + R_c \left(\Omega^2 \left(\frac{1}{2} I_0^c - \frac{2 I_2^c}{h_c^2} \right) - 2 A_{11}^c \alpha_5 \right) (v^b) \\
 & + 2 \Omega (2 R_t I_0^t + R_c I_0^c) \left(\frac{\partial w}{\partial t} \right) + R_c \left(\frac{2 I_2^c}{h_c^2} - \frac{I_0^c}{2} \right) \left(\frac{\partial^2 v^b}{\partial t^2} \right) + 2 \frac{A_{22}^t}{R_t} \left(\frac{\partial^2 v^t}{\partial \theta^2} \right) \\
 & - \left(2 R_t \left(\frac{a_1}{2\pi R_t} q_f + a_2 N_{T\theta}^t - a_3 \sigma_r h_t - A_{33}^t \right) + \frac{1}{2} R_c \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c - a_3 \sigma_r h_c \right) \right) \left(\frac{\partial^2 v^t}{\partial x^2} \right) \\
 & - \frac{1}{2} R_c \left[\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right] \left(\frac{\partial^2 v^b}{\partial x^2} \right) \\
 & + \left(2 \left(\frac{A_{22}^t}{R_t} - R_c A_{11}^c \alpha_2 \alpha_3 \right) + R_c \Omega^2 \left(\left(\frac{I_0^c}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) + \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) \right) \left(\frac{\partial w}{\partial \theta} \right) \\
 & + \left(\Omega^2 \left(2 R_t I_0^t + \frac{1}{2} R_c I_0^c + \frac{2 R_c I_2^c}{h_c^2} \right) - 2 R_c A_{11}^c \alpha_3^2 \right) (v^t) \\
 \delta v^b : & \left(-\frac{1}{4} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left[\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right] - 2 (B_{12}^b + 2 B_{33}^b) \right) \left(\frac{\partial^3 w}{\partial x^2 \partial \theta} \right) \\
 & + R_c \left(-\left(\frac{I_0^c}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) + \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) \left(\frac{\partial^3 w}{\partial \theta \partial t^2} \right) - 2 \frac{B_{22}^b}{R_b^2} \left(\frac{\partial^3 w}{\partial \theta^3} \right) \\
 & + 2 (A_{12}^b + A_{33}^b) \left(\frac{\partial^2 u^b}{\partial x \partial \theta} \right) - \frac{1}{2} R_c \left(a_3 \sigma_r h_c + \frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c \right) \left(\frac{\partial^2 v^t}{\partial x^2} \right) \\
 & + R_c \left(2 \frac{I_2^c}{h_c^2} - \frac{I_0^c}{2} \right) \left(\frac{\partial^2 v^t}{\partial t^2} \right) - \left(2 R_b I_0^b + \frac{1}{2} R_c I_0^c + \frac{2 R_c I_2^c}{h_c^2} \right) \left(\frac{\partial^2 v^b}{\partial t^2} \right) \\
 & + 2 \frac{A_{22}^b}{R_b} \left(\frac{\partial^2 v^b}{\partial \theta^2} \right) - \left(2 R_b \left(\frac{a_1}{2\pi R_b} q_f + a_2 N_{T\theta}^b - a_3 \sigma_r h_b - A_{33}^b \right) + \frac{1}{2} R_c \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) \right) \left(\frac{\partial^2 v^b}{\partial x^2} \right) \\
 & + R_c \left(\Omega^2 \left(\frac{1}{2} I_0^c - 2 I_2^c \frac{1}{h_c^2} \right) - 2 A_{11}^c \alpha_5 \right) (v^t) + \left(2 \left(\frac{A_{22}^b}{R_b} + R_c A_{11}^c \alpha_2 \alpha_4 \right) + R_c \Omega^2 \left(\frac{I_0^c}{4} \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) - \frac{I_2^c}{h_c^2} \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) \right) \left(\frac{\partial w}{\partial \theta} \right) \\
 & + 2 \Omega (2 R_b I_0^b + R_c I_0^c) \left(\frac{\partial w}{\partial t} \right) + \left(\Omega^2 \left(2 R_b I_0^b + \frac{1}{2} R_c I_0^c + 2 R_c I_2^c \frac{1}{h_c^2} \right) - 2 R_c A_{11}^c \alpha_4^2 \right) (v^b) \\
 \delta w : & \left(2 R_t I_2^t + 2 R_b I_2^b + \frac{1}{8} R_c I_0^c (h_t - h_b)^2 + \frac{1}{2} R_c I_2^c \left(\frac{1}{h_c} \right)^2 (h_t + h_b)^2 \right) \left(\frac{\partial^4 w}{\partial x^2 \partial t^2} \right) \\
 & + 2 \left(\frac{B_{12}^t}{R_t} + 2 \frac{B_{33}^t}{R_t} \right) \left(\frac{\partial^3 u^t}{\partial x \partial \theta^2} \right) + 2 R_t B_{11}^t \left(\frac{\partial^3 u^t}{\partial x^3} \right) + 2 R_b B_{11}^b \left(\frac{\partial^3 u^b}{\partial x^3} \right) \\
 & + R_c \left(\frac{1}{4} (h_t - h_b) I_0^c + I_2^c \left(\frac{h_t + h_b}{h_c^2} \right) \right) \left(\frac{\partial^3 u^t}{\partial x \partial t^2} \right) \\
 & + \left(2 \frac{I_2^t}{R_t} + 2 \frac{I_2^b}{R_b} + \frac{1}{8} R_c I_0^c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 + \frac{1}{2} R_c I_2^c \left(\frac{1}{h_c} \right)^2 \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right)^2 \right) \left(\frac{\partial^4 w}{\partial \theta^2 \partial t^2} \right)
 \end{aligned}$$

$$\begin{aligned}
& -2 \left(\frac{D'_{22}}{R_t^3} + \frac{D^b_{22}}{R_b^3} \right) \left(\frac{\partial^4 w}{\partial \theta^4} \right) - 2 \left(R_t D'_{11} + R_b D^b_{11} \right) \left(\frac{\partial^4 w}{\partial x^4} \right) + 2 \frac{B'_{22}}{R_t^2} \left(\frac{\partial^3 v^t}{\partial \theta^3} \right) \\
& + 2 \left(\frac{B^b_{12}}{R_b} + 2 \frac{B^b_{33}}{R_b} \right) \left(\frac{\partial^3 u^b}{\partial x \partial \theta^2} \right) \\
& - \left(\frac{4}{R_t} (D'_{12} + 2D'_{33}) + \frac{4}{R_b} (D^b_{12} + 2D^b_{33}) + \frac{1}{8} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c - a_3 \sigma_r h_c \right) \right) \left(\frac{\partial^4 w}{\partial x^2 \partial \theta^2} \right) \\
& + R_c \left(\left(\frac{h_t - h_b}{4} \right) I_0^c - \left(\frac{h_t + h_b}{h_c^2} \right) I_2^c \right) \left(\frac{\partial^3 u^b}{\partial x \partial t^2} \right) + 2 \frac{B^b_{22}}{R_b^2} \left(\frac{\partial^3 v^b}{\partial \theta^3} \right) \\
& + \left(2(B'_{12} + 2B'_{33}) + \frac{1}{4} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) \right) \left(\frac{\partial^3 v^t}{\partial x^2 \partial \theta} \right) \\
& + R_c \left(\left(\frac{I_0^c}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) + \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) \left(\frac{\partial^3 v^t}{\partial \theta \partial t^2} \right) + R_c \left(\left(\frac{I_0^c}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) - \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) \left(\frac{\partial^3 v^b}{\partial \theta \partial t^2} \right) \\
& + \left(2(B^b_{12} + 2B^b_{33}) + \frac{1}{4} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) \right) \left(\frac{\partial^3 v^b}{\partial x^2 \partial \theta} \right) \\
& - 2 \left(R_t I_0^t + R_b I_0^b + R_c I_0^c \right) \left(\frac{\partial^2 w}{\partial t^2} \right) - 2\Omega R_c I_0^c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{\partial^2 w}{\partial t \partial \theta} \right) - 2\Omega (2R_t I_0^t + R_c I_0^c) \left(\frac{\partial v^t}{\partial t} \right) \\
& + \left(-\Omega^2 \left(2 \frac{I_0^t}{R_t} + 2 \frac{I_0^b}{R_b} + \frac{1}{8} R_c I_0^c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right)^2 + \frac{1}{2} R_c I_2^c \frac{1}{h_c^2} \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right)^2 \right) + 4 \frac{B'_{22}}{R_t^2} + 4 \frac{B^b_{22}}{R_b^2} + 2R_c A_{11}^c \alpha_2^2 \right) \left(\frac{\partial^2 w}{\partial \theta^2} \right) \\
& + 2 \left(-A'_{12} + \left(\frac{1}{h_c} \right) \alpha_1 R_c A_{11}^c \right) \left(\frac{\partial u^t}{\partial x} \right) + 2 \left(-A^b_{12} + \left(\frac{1}{h_c} \right) \alpha_1 R_c A_{11}^c \right) \left(\frac{\partial u^b}{\partial x} \right) \\
& - 2 \left(\frac{q_f}{2\pi} - \sigma_r [R_b h_b + R_c h_c + R_t h_t] + R_t N_{Tx}^t + R_c N_{Tx}^c + R_b N_{Tx}^b - 2B'_{12} - 2B^b_{12} - R_c A_{11}^c \alpha_1^2 \right) \left(\frac{\partial^2 w}{\partial x^2} \right) \\
& + \left(2R_b I_0^b + R_c I_0^c \right) \left(\frac{\partial v^b}{\partial t} \right) + 2 \left(\Omega^2 \left(R_t I_0^t + R_b I_0^b + R_c I_0^c \right) - \left(\frac{A'_{22}}{R_t} + \frac{A^b_{22}}{R_b} \right) \right) (w) \\
& + \left(-R_c \Omega^2 \left(\left(\frac{I_0^c}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) + \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) + 2 \left(-\frac{A'_{22}}{R_t} + R_c A_{11}^c \alpha_2 \alpha_3 \right) \right) \left(\frac{\partial v^t}{\partial \theta} \right) \\
& - \left(R_c \Omega^2 \left(\left(\frac{I_0^c}{4} \right) \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) - \left(\frac{I_2^c}{h_c^2} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \right) + 2 \left(\frac{A^b_{22}}{R_b} + R_c A_{11}^c \alpha_2 \alpha_4 \right) \right) \left(\frac{\partial v^b}{\partial \theta} \right) \tag{29}
\end{aligned}$$

So, applying the effect of B.C.'s, the final equations are obtained:

$$\begin{aligned}
\delta u^t : & 2R_t B'_{11} \left(\frac{\partial^2 w}{\partial x^2} \right) + 2 \frac{B'_{12}}{R_t} \left(\frac{\partial^2 w}{\partial \theta^2} \right) - 2R_t A'_{11} \left(\frac{\partial u^t}{\partial x} \right) - 2A'_{12} \left(\frac{\partial v^t}{\partial \theta} \right) - 2A'_{12}(w) \\
\delta u^b : & 2R_b B^b_{11} \left(\frac{\partial^2 w}{\partial x^2} \right) + 2 \frac{B^b_{12}}{R_b} \left(\frac{\partial^2 w}{\partial \theta^2} \right) - 2R_b A^b_{11} \left(\frac{\partial u^b}{\partial x} \right) - 2A^b_{12} \left(\frac{\partial v^b}{\partial \theta} \right) - 2A^b_{12}(w) \\
\delta v^t : & \left(\frac{1}{4} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) + 4B'_{33} \right) \left(\frac{\partial^2 w}{\partial x \partial \theta} \right) \\
& + \frac{1}{2} R_c \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) \left(\frac{\partial v^b}{\partial x} \right) - 2A'_{33} \left(\frac{\partial u^t}{\partial \theta} \right) \\
& + \left(2R_t \left(\frac{a_1}{2\pi R_t} q_f + a_2 N_{T\theta}^t - a_3 \sigma_r h_t - A'_{33} \right) + \frac{1}{2} R_c \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c - a_3 \sigma_r h_c \right) \right) \left(\frac{\partial v^t}{\partial x} \right) \\
\delta v^b : & \left(\frac{1}{4} R_c \left(\frac{h_t}{R_t} - \frac{h_b}{R_b} \right) \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) + 4B^b_{33} \right) \left(\frac{\partial^2 w}{\partial x \partial \theta} \right) \\
& + \frac{1}{2} R_c \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) \left(\frac{\partial v^t}{\partial x} \right) - 2A^b_{33} \left(\frac{\partial u^b}{\partial \theta} \right) \\
& + \left(2R_b \left(\frac{a_1}{2\pi R_b} q_f + a_2 N_{T\theta}^b - a_3 \sigma_r h_b - A^b_{33} \right) + \frac{1}{2} R_c \left(\frac{a_1}{2\pi R_c} q_f + a_2 N_{T\theta}^c + a_3 \sigma_r h_c \right) \right) \left(\frac{\partial v^b}{\partial x} \right)
\end{aligned}$$

$$\delta\left(\frac{\partial w}{\partial x}\right) : -2\left(R_t D_{11}^t + R_b D_{11}^b\right)\left(\frac{\partial^2 w}{\partial x^2}\right) - 2\left(\frac{D_{12}^t}{R_t} + \frac{D_{12}^b}{R_b}\right)\left(\frac{\partial^2 w}{\partial \theta^2}\right) + 2R_t B_{11}^t\left(\frac{\partial u^t}{\partial x}\right) + 2R_b B_{11}^b\left(\frac{\partial u^b}{\partial x}\right) + 2B_{12}^t\left(\frac{\partial v^t}{\partial \theta}\right) + 2B_{12}^b\left(\frac{\partial v^b}{\partial \theta}\right) + 2\left(B_{12}^t + B_{12}^b\right)(w) \quad (30)$$

The coefficients α_i ($i = 1, 2, \dots, 5$) are stated in Appendix A2.

The boundary conditions at two ends of the sandwich cylindrical shell are considered as follows:

for clamped boundary conditions $u^i = v^i = w = \phi_x^c = \phi_\theta^c = 0$

for simply supported (roller) boundary conditions $w = \frac{\partial u^i}{\partial x} = \frac{\partial v^i}{\partial x} = \frac{\partial \phi_\theta^c}{\partial \theta} = \frac{\partial \phi_x^c}{\partial x} = \frac{\partial^2 w}{\partial x^2} = \frac{\partial^2 w}{\partial \theta^2} = 0$

for simply supported (hinged) boundary conditions $u^i = v^i = w = \frac{\partial \phi_\theta^c}{\partial \theta} = \frac{\partial \phi_x^c}{\partial x} = \frac{\partial^2 w}{\partial x^2} = \frac{\partial^2 w}{\partial \theta^2} = 0$

4 Solution and validation method

4.1 Differential quadrature method (DQM)

The use of analytical methods in solving engineering problems is always preferable to other methods due to the fact that they are highly accurate. But today, with the expansion of science and the problems become more complex, it may not be possible to use analytical methods. In addition, the use of analytical methods in some problems is associated with limitations, so numerical methods in the analysis of engineering problems replace analytical methods. Low-order numerical methods such as FEM and FDM increase the number of computational nodes for sufficient accuracy. In higher-order methods such as differential quadratic, even with few computing nodes, the numerical results of this method are accurate. Today, with the spread of science and problems becoming more complex, it may not be possible to use analytical methods that have very high accuracy and are always preferable to other methods, or these methods are associated with limitations in solving problems. As a result, this has led to the increasing use of numerical methods in analyzing engineering problems. The finite element method and finite difference are part of low-order numerical methods, which means that in order to have sufficient accuracy with employing these methods, a large number of computational nodes should be considered. The differential quadratic method (DQM) is known as a high-order method, and even with the use of a small number of computing nodes, the numerical results of this method have good accuracy. The method of difference of differential squares is one of the numerical methods in which the governing differential equations are converted into a group of first-order algebraic equations using the weight coefficients. Thus, the derivative will be expressed at each point as a linear sum of weight coefficients and function values at that point and other domain points and in the direction of the coordinate axes. This method can solve all kinds of equations and a set of coupled equations. The differential quadrature method is used to solve the governing equations of motion for the rotating visco-elastic sandwich beam and obtain the structure's natural frequencies. Using the weight coefficient matrix, this method converts the governing equations of motion into first-order algebraic equations. The selection of sample points and weighting coefficients are important factors in the accuracy of the differential square method. The best distribution of sample points is such that the density of considered points near the border areas of the domain is higher than the middle areas of the domain. In this case, the function is approximated at the boundaries with more points, and the accuracy of the analysis increases. One of the most important points in the differential quadratic method is the application of boundary conditions. In this method, boundary conditions are applied through weight matrices. If this method is used, the obtained answers will be very accurate, and the number of points required to achieve convergence of the problem will be reduced. This research uses the differential quadratic method to solve the governing equations of rotating nanocomposite sandwich shells. The idea and basis of the differential quadratic method is to use the Lagrange approximation. In this way, the governing differential equations are converted into a group of algebraic equations by using weight coefficients. Thus, the derivative is expressed at each point as a linear sum of weight coefficients and function values at that point and other domain points and in the direction of the coordinate axes. In this research, both the exact solution method and the differential quadratic method are used to obtain the forward and backward frequencies of the rotating sandwich cylindrical shell, and finally, to validate the equations and the solution method presented, a comparison between both methods is presented,

and the other researchers are done. Based on the differential square method, for an arbitrary function, the derivative of order r at an arbitrary point i is defined as follows [58]:

$$\frac{d^m}{dx^m} f(x_i) = \sum_{j=1}^N A_{ij}^{(m)} f(x_j) \quad i = 1, 2, \dots, N \quad (31)$$

$f(x_j)$ represents the value of the desired function at the x_i desired point. Also, $A_{ij}^{(m)}$ is the matrix of weight coefficients corresponding to the derivative of order r , which is obtained as:

$$A_{ij}^{(1)} = \begin{cases} \prod_{\substack{m=1 \\ m \neq i, j}}^N (x_i - x_m), & (i, j = 1, 2, 3, \dots, N; i \neq j) \\ \prod_{\substack{m=1 \\ m \neq j}}^N (x_j - x_m) & A^{(r)} = A^{(r-1)} A^{(1)} \quad 2 \leq r \leq N-1 \\ \sum_{\substack{m=1 \\ m \neq i}}^N \frac{1}{(x_i - x_m)}, & (i = j = 1, 2, 3, \dots, N) \end{cases} \quad (32)$$

In this method, the derivatives of the function are considered only at certain points of the domain, so the solutions become more accurate as the points increase, and this increase in accuracy continues until the convergence point of the solution. In addition to the number of points, how the points are distributed in the domain of problem-solving is also important. One of the most important points in the differential quadratic method is the application of boundary conditions. There are different ways to apply boundary conditions, each with advantages and limitations. In the method of applying boundary conditions through weight matrices, if there is a limit on each of the derivative orders in the boundaries of the problem, this limit shows itself in the form of shear locking phenomenon. If it is possible to use this method, the answers obtained will be very accurate, and the number of points required for the convergence of the problem will be very small. The biggest problem of the other method is its inability to solve problems with combined boundary conditions. To solve this problem, the method of separating the problem interval into domain and boundary points is used by considering DQM. In this method, the points in the problem-solving range are divided into two categories: domain and boundary points. The differential relation governing the problem is written only for the domain points, and the boundary conditions are written separately for the boundary points. Then, the two obtained relations are combined with each other, and the final relation for the problem is calculated. This method can solve the problem for different boundary conditions, including combined or simply supported (hinged or roller) boundary conditions, and no restrictions are created for this method. In this research, the cosine distribution known as the Chebyshev–Gauss–Lobatto distribution has used. The advantage of this type of point distribution is that the density of the considered points near to two end points (boundary conditions) is higher than the middle points; and therefore, the function is approximated by more points at the points where sudden changes are applied to the boundaries, and the accuracy of the problem-solving increase. One of the best point distributions that have considered so far is the cosine distribution known as the Chebyshev–Gauss–Lobatto distribution, which is expressed as follows [58]:

$$x_i = \frac{1}{2}L \left\{ 1 - \cos \left[\frac{(i-1)\pi}{(N-1)} \right] \right\} \quad i = 1, 2, \dots, N \quad (33)$$

Finally, the derivatives of the functions of the equation governing the rotary nanocomposite sandwich shell are obtained:

$$\begin{aligned} \frac{df}{dx} &= [A^{(1)}]\{f\}, \quad \frac{d^2f}{dx^2} = [A^{(2)}]\{f\}, \quad \frac{d^3f}{dx^3} = [A^{(3)}]\{f\}, \quad \frac{d^4f}{dx^4} = [A^{(4)}]\{f\} \\ u^i(x, \theta, t) &= [U_{si}(x) \cos(n\theta) + U_{pi}(x) \sin(n\theta)] \exp(i\omega t) \\ v^i(x, \theta, t) &= [V_{si}(x) \cos(n\theta) + V_{pi}(x) \sin(n\theta)] \exp(i\omega t) \quad i = t, b \quad s = 1 \quad p = 2 \\ w(x, \theta, t) &= [W_s(x) \cos(n\theta) + W_p(x) \sin(n\theta)] \exp(i\omega t) \end{aligned} \quad (34)$$

ω , n , $U_{si}(x)$, $V_{si}(x)$ and $W_s(x)$ represents the frequency, the number of environmental modes, and the range of vibrations of the rotating nanocomposite sandwich shell in three longitudinal, circumferential, and

Table 4 Mechanical properties and geometric characteristics of sandwich cylindrical shell with simple support

Mechanical properties and geometrical characteristics	
$R_b = 100 \text{ mm}, R_c = \left[100 + \frac{h_b + h_c}{2}\right] \text{ mm}, R_t = \left[100 + \frac{h_b + 2h_c + h_t}{2}\right] \text{ mm}, E_t = E_b = 210 \text{ Gpa}, h_t = h_b = h_c = 1/3 \text{ mm}$	
$\nu_t = \nu_b = 0.3, \rho_t = \rho_b = 7850 \text{ kg/m}^3, \rho_c = 1340 \text{ kg/m}^3, G_c = 8.582 [1 + 0.3478i] \text{ Mpa}, L = 100 \text{ mm}$	

transverse directions, respectively. In the vibration of a rotating nanocomposite sandwich shell, all terms with time derivative in the governing equation are rewritten in terms of spatial derivative.

By placing Eq. (34) in the equations of the rotary nanocomposite sandwich shell, then the algebraic form of these equations is obtained as follows:

$$\begin{aligned} \{X\} + \omega[G]\{X\} + \omega^2[M]\{X\} &= 0 \\ \{X\} &= [\{U_{si}\}, \{U_{pi}\}, \{V_{si}\}, \{V_{pi}\}, \{W\}]^T \quad i = t, b \quad s = 1 \quad p = 2 \end{aligned} \quad (35)$$

$[K]$, $[G]$ and $[M]$ represents the stiffness matrix, gyroscopic effects, and mass matrix of rotating nanocomposite sandwich shell, respectively.

4.2 Validation

In this research, Navier's exact solution method, as well as the differential quadratic method, are used to obtain the frequencies of the rotational nanocomposite sandwich shell due to its stability and high efficiency. In the differential quadratic method, choosing the number of grid points is very important in the accuracy of the obtained answers. It is also necessary to validate the provided equations and solution methods before presenting the obtained results. For this purpose, in the first step, a homogeneous shell with a visco-elastic core and a simple support along two edges without considering the effect of residual stresses, thermomechanical loading, and rotational speed to check the convergence of the differential quadratic method and also validate the equations and the exact solution method is used. The geometric characteristics and mechanical properties of sandwich cylindrical shells are shown in Table 4.

In this article, the formulas for all visco-elastic spinning systems FG graphene platelets reinforced nanocomposite sandwich cylindrical shell with MRE core have obtained and solved. These formulas can be used for different structure dimensions, composite matrix material, and nano reinforcement. Also, the structure's core can include various types mentioned in Table 1. However, to verify and compare the effect of different parameters on the overall frequency of the structure, certain geometric features and mechanical properties are needed for the sandwich cylindrical shell, as shown in Table 4. The obtained results are compared to the other research based on geometric and mechanical properties in Table 4 that these validations are considered in Tables 5 and 6. By changing each of these parameters, the frequency of the structure changes according to whether this parameter decreases or increases the structure's stiffness.

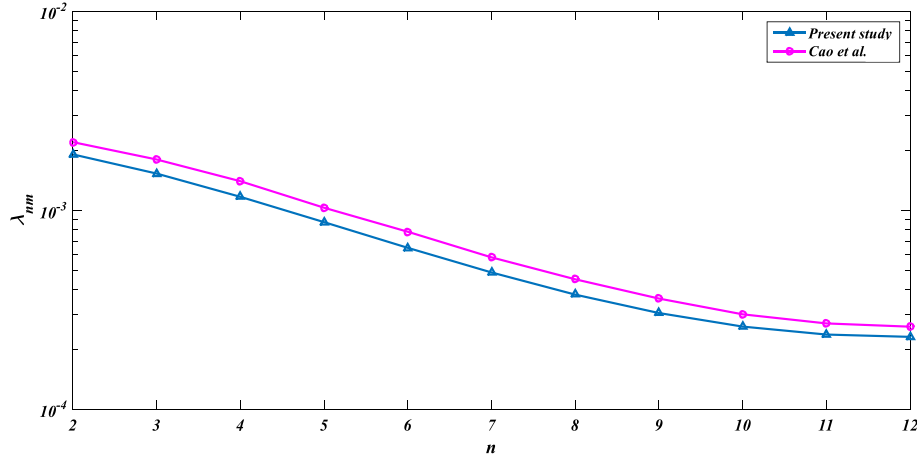
In Table 5, the convergence of the frequencies for the sandwich shell and the number of optimal points based on the differential quadratic method is compared with Song et al. [60] and Xiang et al. [61]. By increasing N ,

Table 5 Natural frequencies of cylindrical shell for simply supported B.C.'s and circumferential mode based on the axis symmetrical case ($n = 0$)

Number of grid points (N)	Longitudinal modes (m) ($\times 10^9$)									
	1	2	3	4	5	6	7	8	9	10
DQM ($N = 15$)	2.4580	2.4831	2.5100	2.5653	2.6722	2.8309	3.0313	3.9160	4.4155	5.5440
DQM ($N = 20$)	2.4580	2.4831	2.5100	2.5652	2.6705	2.8527	3.1446	3.5743	4.1550	5.6102
DQM ($N = 25$)	2.4580	2.4831	2.5100	2.5651	2.4705	2.8527	3.1440	3.5827	4.2120	5.0788
DQM ($N = 30$)	2.4580	2.4831	2.5100	2.5651	2.6705	2.8527	3.1440	3.5827	4.2125	5.0833
DQM ($N = 35$)	2.4580	2.4831	2.5100	2.5651	2.6705	2.8527	3.1440	3.5827	4.2125	5.0833
Navier's solution	2.4599	2.4836	2.5104	2.5657	2.6712	2.8537	3.1455	3.5849	4.2157	5.0881
Song et al. [60]	2.4538	2.4719	2.4992	2.5497	2.6769	2.8211	3.2303	3.6124	4.2267	—
Xiang et al. [61]	2.4438	2.4703	2.4989	2.5566	2.6651	2.8511	3.1470	3.5910	4.2270	5.1050

Table 6 Mixed eigenvalues of sandwich cylindrical shell with visco-elastic core and simply supported B.C's for $n = 0$

m	Present work (DQM)	Present work (Navier)	Shakuri et al. [62]	Permoon et al. [63]
8	$3.58 \times 10^9 + i46.69 \times 10^6$	$3.58 \times 10^9 + i46.69 \times 10^6$	$3.57 \times 10^9 + i43.36 \times 10^6$	$3.57 \times 10^9 + i43.28 \times 10^6$
9	$4.21 \times 10^9 + i56.47 \times 10^6$	$4.21 \times 10^9 + i56.47 \times 10^6$	$4.21 \times 10^9 + i55.12 \times 10^6$	$4.21 \times 10^9 + i55.04 \times 10^6$
10	$5.08 \times 10^9 + i67.40 \times 10^6$	$5.08 \times 10^9 + i67.40 \times 10^6$	$5.08 \times 10^9 + i68.26 \times 10^6$	$5.08 \times 10^9 + i68.17 \times 10^6$

**Fig. 4** Comparison of sandwich shell frequencies with simple support and different circumferential modes with the results of other research

the frequencies of the sandwich shell obtained by the differential quadratic method converge quickly and agree with the values obtained by Navier's exact solution method with completely acceptable accuracy. Therefore, taking into account the number of grid points in addition to the acceptable accuracy of the obtained frequencies will reduce the number of calculations. It also shows that the frequencies of the sandwich structure obtained from the present study based on both DQM and Navier's exact solution method are in good agreement with each other [60, 61].

In another comparison in Table 6, the mixed eigenvalues of the shell with visco-elastic core and simply supported for $n = 0$ and longitudinal modes m are different from the results of Shakuri et al. [62] and Permoon et al. [63] are reviewed. They have a good agreement with each other.

In Tables 5 and 6 and Fig. 2, the obtained results are presented for simply supported boundary conditions (B.C.'s) and circumferential mode based on the axis symmetrical case ($n = 0$). In Tables 5 and 6, only the effect of the axial mode (various m) for the axis symmetrical case ($n = 0$) on the natural frequency is considered; while in Fig. 2 the effect of the circumferential mode (various n) on the natural frequency is shown, because the validation of MATLAB code for two modes (m and n) is investigated.

The frequencies from the present equations and results obtained from Kao et al. [64] for a sandwich shell with simply supported and different values of environmental modes are shown in Fig. 4. As can be seen, there is a good agreement for the sandwich shell with simply supported boundary conditions.

In this research, in order to find the most optimal number of points and solve the equations by the semi-analytical DQM method and compare it with Navier's analytical method, it is first assumed that the boundary conditions are considered as simply supported. First, in the case where the circumferential mode ($n = 0$) based on the axis symmetrical mode is considered constant, then by using the DQM solution method and increasing the number of different points by increasing the number of grid points, the natural frequencies of the sandwich shell obtained by the differential square method are quickly converged and a completely acceptable accuracy is consistent with the values obtained by Navier's exact solution method. Then the number of longitudinal modes m and circumferential mode n are compared. In the authors' previous works, the number of grid points for other states of boundary conditions has investigated. It should also be noted that based on the DQM method, the solution domain is discretized into N separate nodes. At each node, the displacement derivatives are estimated with the weighted coefficients of the displacement value and the derivative of the potential function with the weighted coefficients of the potential value throughout the structure. Also, the effects of boundary conditions are considered by modifying weight coefficients. You should note that in this method, the function's derivatives

are considered only at certain points of the problem-solving domain, so by increasing these points, we can expect that the accuracy of the answers will improve. Of course, this increase must continue until a point of convergence of the solution, and after that this number of points can be used for all modes. In this method, the points in the problem-solving range are divided into two categories: domain and boundary points. The differential relation governing the problem is written only for the domain points, and the boundary conditions are written separately for the boundary points. Then, the two obtained relations are combined, and the problem's final relation will be extracted and any boundary condition can be applied according to the weight functions in the form of said matrices.

5 Numerical results

Residual stresses are self-balanced stresses that can be created in most mechanical structures due to plastic deformation, temperature changes and changes in material structure, and depending on the application of the structure, it may be favorable or unfavorable. The effect of residual stresses in the structure is as important as mechanical stresses, and considering them can have an important effect on the behavior of mechanical structures. The strain energy caused by the residual stresses in the inner and outer layers of the rotary nanocomposite sandwich cylindrical shell can appear as a deviation in the structure's overall structure. In this research, the residual stresses effect is considered a symmetrical deviation in the z-direction, and the torsional deviation of the cylindrical sandwich shell is also ignored. The effect of tensile and compressive residual stresses on the frequency and loss factor coefficient of the rotating nanocomposite sandwich shell reinforced with uniform distribution of graphene and magnetorheological core is shown in Fig. 5. When the rotating sandwich shell is subjected to tensile residual stresses, the bending stiffness of the structure increases. As a result, the natural frequency of the rotating sandwich shell increases. The loss factor of the structure is also increased due to the increase in the natural frequency and the decrease in the real part of the eigenvalues. As a result, the rotating sandwich shell becomes unstable at higher rotating speeds. It is also observed that with the increase in compressive residual stress in the rotating sandwich shell, the bending stiffness of the structure decreases. As a result, the natural frequency of the shell decreases. The loss factor of the structure is also reduced due to the reduction of the natural frequency and the increase in the real part of the eigenvalues. As a result, the rotating sandwich shell becomes unstable at lower rotational speeds.

The effect of temperature on the frequency and loss factor of the rotating nanocomposite shell reinforced with a uniform distribution of graphene and magnetorheological core is shown in Fig. 6. In this research, all the material properties of the face sheet layers for the nanocomposite shell have been considered depending on the temperature. Also, the loss coefficient and shear modulus of the magnetorheological core of the sandwich shell are also considered dependent on the temperature and magnetic field because the loss coefficient and shear modulus of the core change under temperature and magnetic field. Increasing the temperature in the magnetorheological core causes the destruction of the hard parts of the core. Therefore, increasing the temperature and applying the magnetic field causes the iron particles in the magnetorheological core to find a regular arrangement along the field, and as a result, the shear modulus for the core is enhanced. As the temperature enhances further due to the excessive softness of the elastomer, the shear modulus gradually decreases from its maximum value at the temperature (353 K). The same thing happens with the loss coefficient. As the temperature increases, the bending stiffness of face sheet layers of the rotating nanocomposite sandwich shell decreases, and considering that its effect is greater than the enhancement in shear modulus of the core due to the enhancement in temperature, finally, temperature increases of rotating nanocomposite sandwich shell leads to a decrease in frequency without. Then, as a result, it leads to the instability of the structure at higher rotational speeds. In the case of the loss factor of the rotating nanocomposite sandwich shell, the behavior is different because the core plays the main role. It is shown that the loss coefficient increases with increasing temperature up to (353 K) and then it starts to decrease as the temperature increases and approaches the temperature (333 K). On the other hands, according to Figs. 2 and 3, with increasing of temperature, the storage shear modulus and loss factor for core and Young's modulus decrease, because the stiffness of a sandwich cylindrical shell reduces, then the natural frequency decreases. The value of constant parameters in Fig. 6 includes $h_b = h_t = 3$ mm, $h_c = 8$ mm, $W_{GPL} = 0.3\%$, $M_f = 0.2$ T, $R_c/H = 50$, $L/R_c = 5$, $\sigma_r = -0.02$.

Magnetorheological materials are a group of smart materials that, due to their rheological properties, are able to change continuously, quickly and reversibly due to the application of a magnetic field. Due to this feature, these materials can be used to control the vibrations of structures. Magnetorheological elastomers are composite structures consisting of magnetic particles and polymers in the role of matrix, which changes

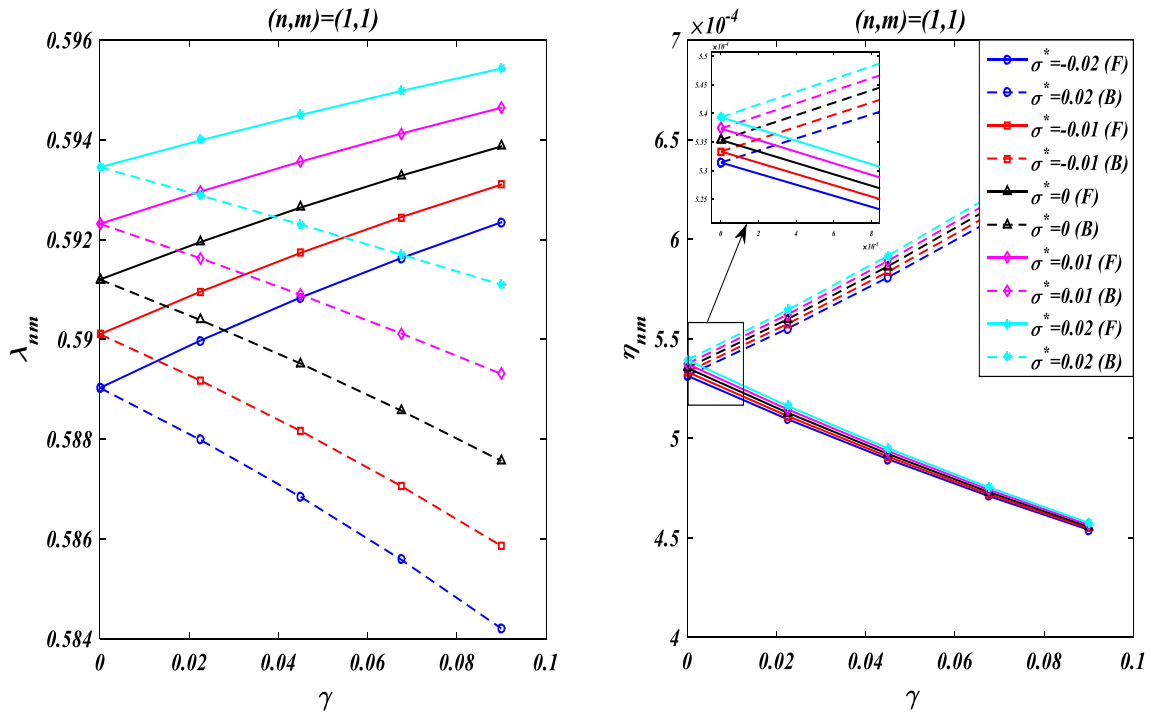


Fig. 5 Effect of tensile and compressive residual stresses on the frequency and loss factor of rotating nanocomposite sandwich shell reinforcement

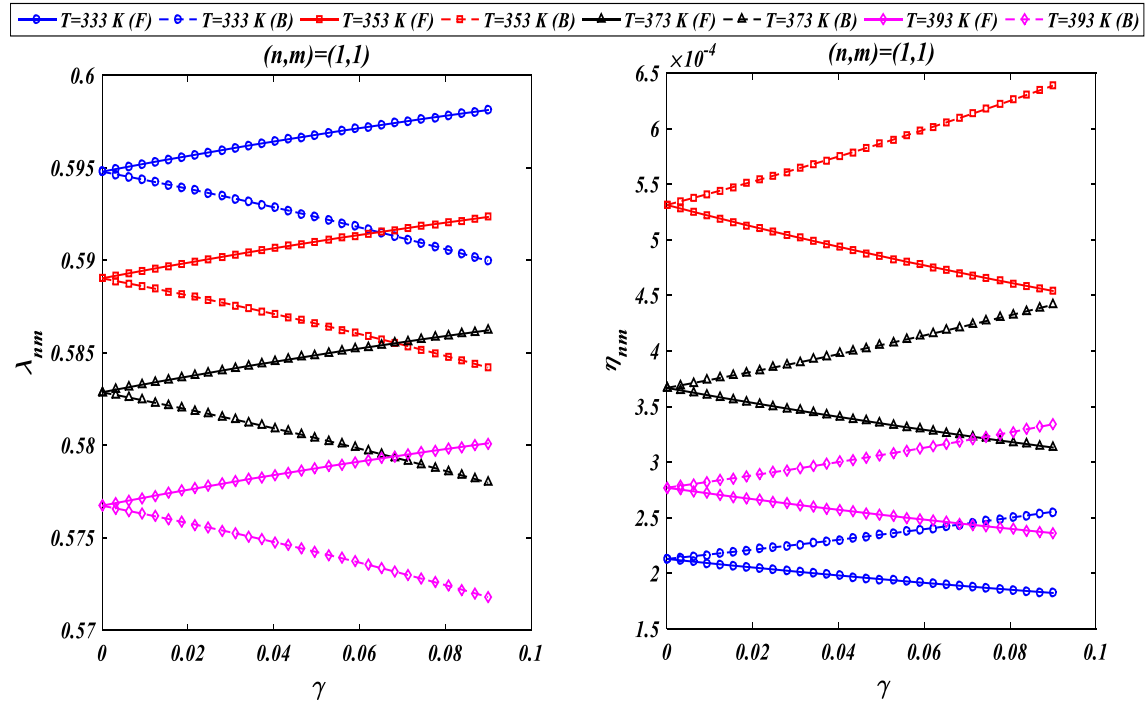


Fig. 6 Effect of temperature on the frequency and loss factor of the rotating nanocomposite sandwich shell reinforced with uniform distribution of graphene and magnetorheological core

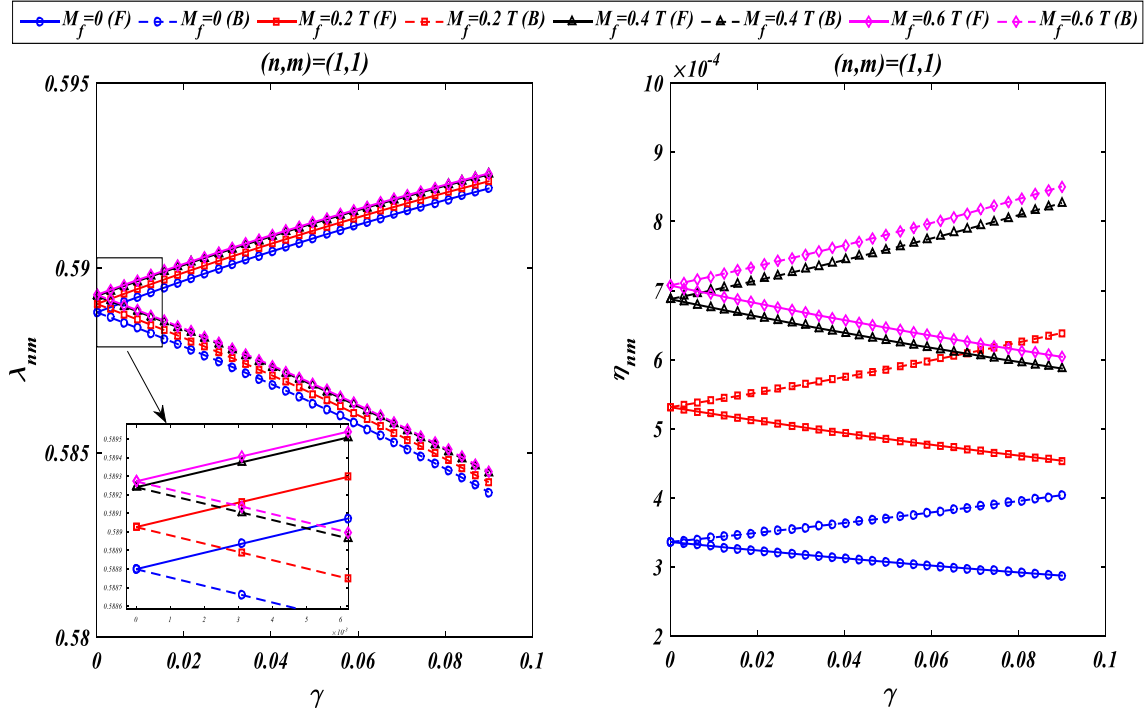


Fig. 7 Effect of magnetic field on loss factor and frequency of rotating nanocomposite sandwich shell reinforced with uniform distribution of graphene and magnetorheological core at $T = 353$

the mechanical properties of the structure due to the movements of these particles under the magnetic field. Magnetorheological elastomers are divided into several categories based on the type of matrix, the most common of which are natural rubber and silicone rubber. Using silicone rubber instead of natural rubber as a matrix improves the mechanical properties of the structure. Other constituent particles, which include magnetic particles (iron particles) and softening percentage, are common in all structures, with the difference that the percentage of these particles can be variable. Magnetic particles play a significant role in improving the performance of magnetorheological elastomers, but it should be noted that the excessive increase in these particles can also be destructive and cause a decrease in the tensile strength of the structure. Considering the assumption considered for the core layer of the rotating sandwich cylindrical shell in the state before yielding, consider the behavior of the linear visco-elastic material for the core. In this case, the relationship between shear modulus and loss coefficient based on natural rubber (60% iron particles, 20% softener and 20% matrix) is strongly dependent on the magnetic field. Also, the Young's modulus of the core is very small compared to the inner and outer layers of the rotating sandwich cylindrical shell, so the normal stress in the core can be ignored. In addition, due to the fact that the magnetic field is applied to the sandwich structure in the z -direction, as a result of the application of the magnetic field, it causes the iron particles in the magnetorheological core to find a regular arrangement in the direction of the field, and as a result, the shear modulus of the core increases. According to Fig. 2, with an increase in the magnetic flux density (Tesla (T)) to 0.6 T, the shear modulus and the loss factor increases, and for higher than, it is approximately constant. Then with increasing the magnetic flux density, the natural frequency increase to 0.6 T. Thus by considering the value of higher 0.6 to 1.2 T, the natural frequency is approximately constant, then in this figure, the effect of M_f to 0.6 T is plotted. The effect of the magnetic field on the loss factor and frequency of the rotating nanocomposite sandwich shell reinforced with a uniform distribution of graphene and magnetorheological core is shown in Fig. 7. The loss factor and shear modulus of the magnetorheological core of the sandwich shell are completely dependent on temperature and magnetic field. Therefore, with the enhancement of the magnetic field, the iron particles in the core are arranged in a regular manner in the direction of the magnetic field, which increases the rigidity of the core and, as a result, enhances the frequencies. This increases the tolerable rotational speed threshold for the rotating sandwich shell. Thus, the optimal value of the magnetic field is in the range of 0.6 T, and increasing it more than this value does not have a noticeable effect on the frequency and loss factor of the rotating sandwich shell.

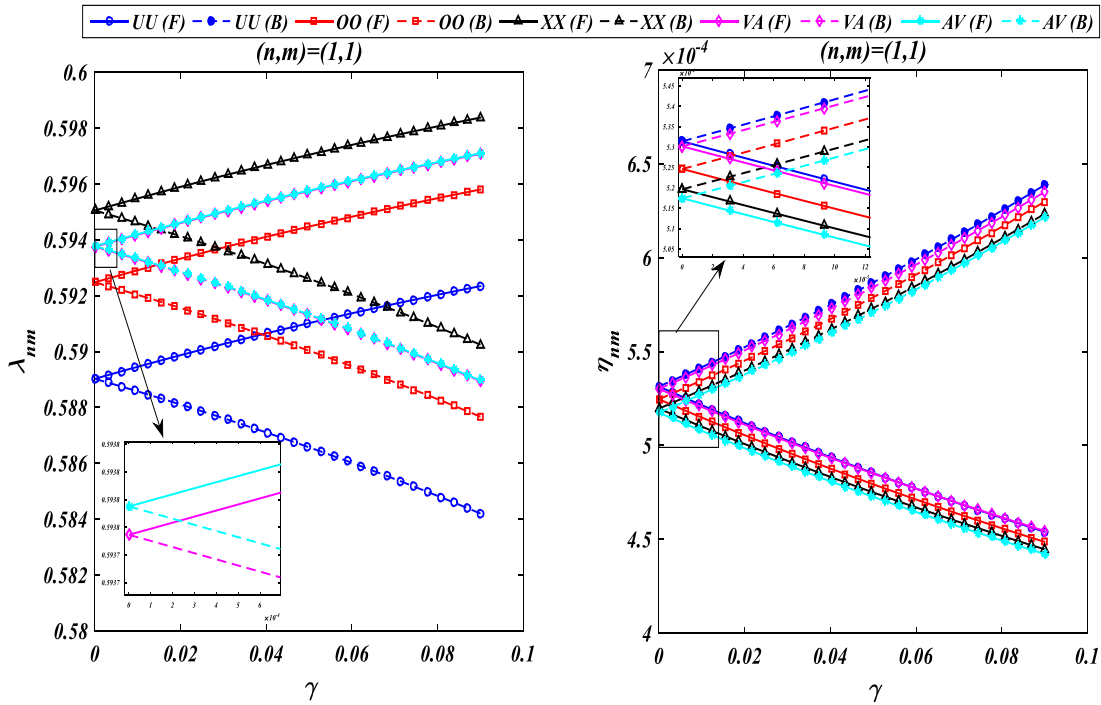


Fig. 8 Effect of different distribution of GPL on loss factor and frequency of rotating nanocomposite sandwich shell

By increasing the amount of nanocomposite matrix reinforcement in places far from the neutral plane, the stiffness and bending tolerance will increase in that area. It will lead to an increase in the natural frequency, as can be seen. In addition to the type of distribution, the values of GPL is more in places far from the core; thus, the natural frequency for that distribution becomes more. Also, if the values of GPL is spread gradually in all areas of procedures, it leads to an increase in frequency. According to Table 2, it can be seen that the two modes, AV and VA, have these characteristics. The effect of different distributions of graphene plates on the frequency and loss factor of a rotating nanocomposite sandwich shell reinforced with graphene and magnetorheological core is shown in Fig. 8. As a result, the lowest frequency of the rotating shell is obtained for the UU distribution GPL, and the highest frequency is also obtained for the XX distribution. The frequencies of the rotating shell in AV and VA distributions are almost the same, although the natural frequency obtained in the AV distribution is slightly higher than in VA.

Controlling the vibrations of rotating structures and preventing destructive phenomena such as flutter is particularly important, especially when the structure is under external loading. To control the vibrations of the rotating structure, the energy input from the external force must be consumed. Considering that the analyzed sandwich structure has a magnetic core and the outer and inner face sheet layers of the structure are also reinforced with GPL, by changing the values of the applied force according to the work caused by axial loading in Eq. (25), the vibrations of the structure can be minimized. The effect of the axial force on the loss factor and the rotating nanocomposite sandwich shell reinforced with uniform distribution of GPL and magnetorheological core are shown in Fig. 9. By enhancement of the axial force on the rotating sandwich shell, the frequency decreases and causes the rotating shell to reach instability at lower rotational speeds. Also, the enhancement in axial force enhances the loss factor of the rotating sandwich shell.

Nanocomposites reinforced with graphene sheets have been expanded in chemical, aerospace, and nuclear reactors that work in environments with temperature gradients. The advantage of using graphene-reinforced metallic nanocomposites is that the thermal conductivity of graphene is low, and it increases the resistance and maintains the structural integrity of these materials in environments with temperature gradients, also considering the properties of carbon and the presence of high attraction between molecules. Adding any of the carbon derivatives to the composite matrix causes a change in the arrangement of the molecules, thus increasing the structure's stiffness as a result of increasing the frequency of the structure. The effect of the volume percentage of graphene platelets (GPLs) on the loss factor and frequency of the rotating nanocomposite sandwich shell reinforced with uniform distribution of GPLs, as well as the magnetorheological core, is shown

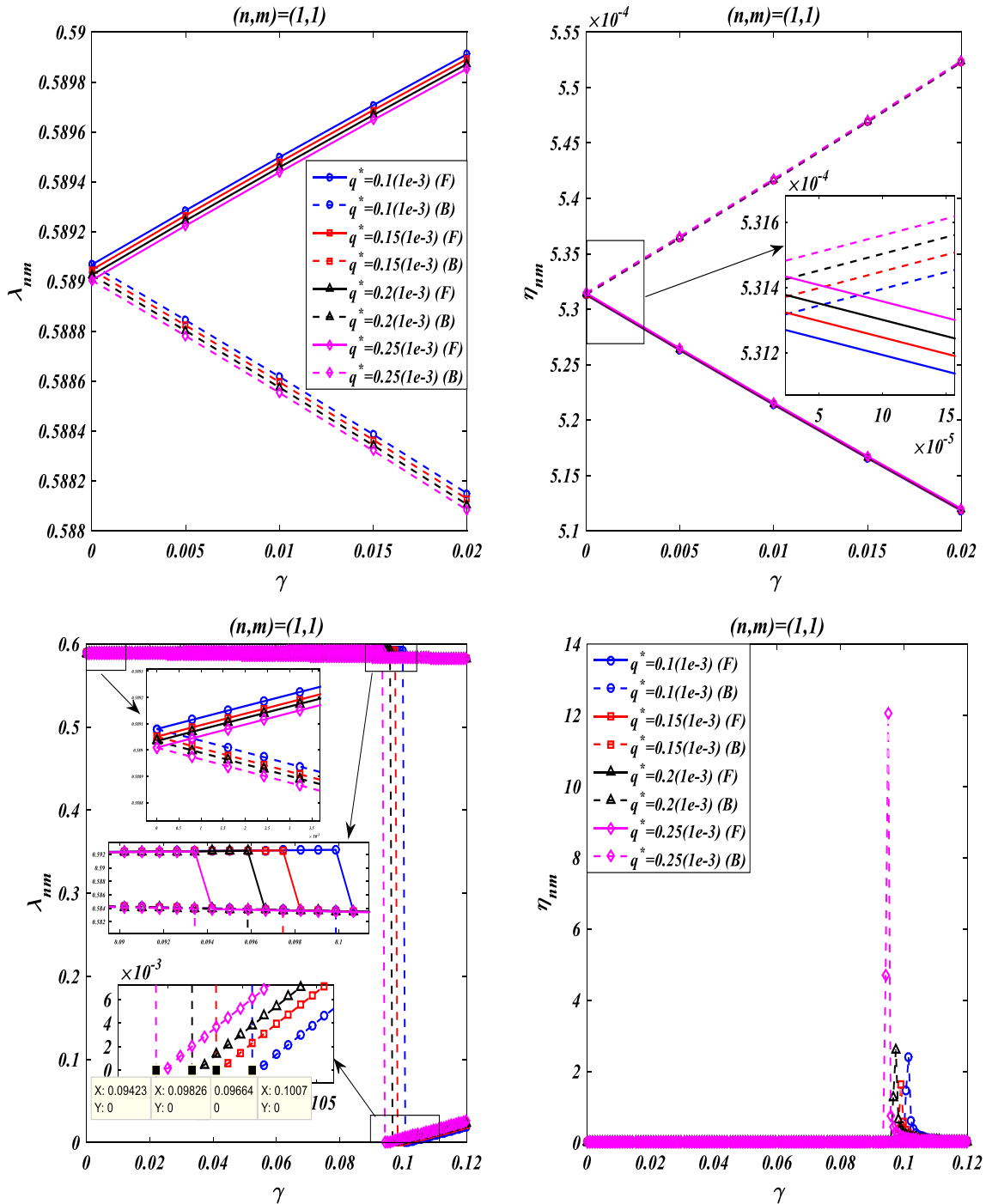


Fig. 9 Effect of axial force on loss factor and frequency of a rotating nanocomposite sandwich shell

in Fig. 10. By increasing the volume percentage of GPLs, the frequency of the rotating shell reduces and causes the instability of the structure at lower rotational speeds. Also, the enhancement of the volume fraction of GPLs enhances the loss factor. The rotating sandwich shell is subjected to thermomechanical loading, so in such structures, one should be very careful about adding GPLs because this increase may increase the modulus and consequently increase the stiffness of the structure, but the properties of the material depend on temperature. It can cause an increase in thermal axial load and, as a result, decrease the frequency and stability range of the structure. In addition, GPL, which is a kind of carbon nanostructure, is used to place carbon particles between

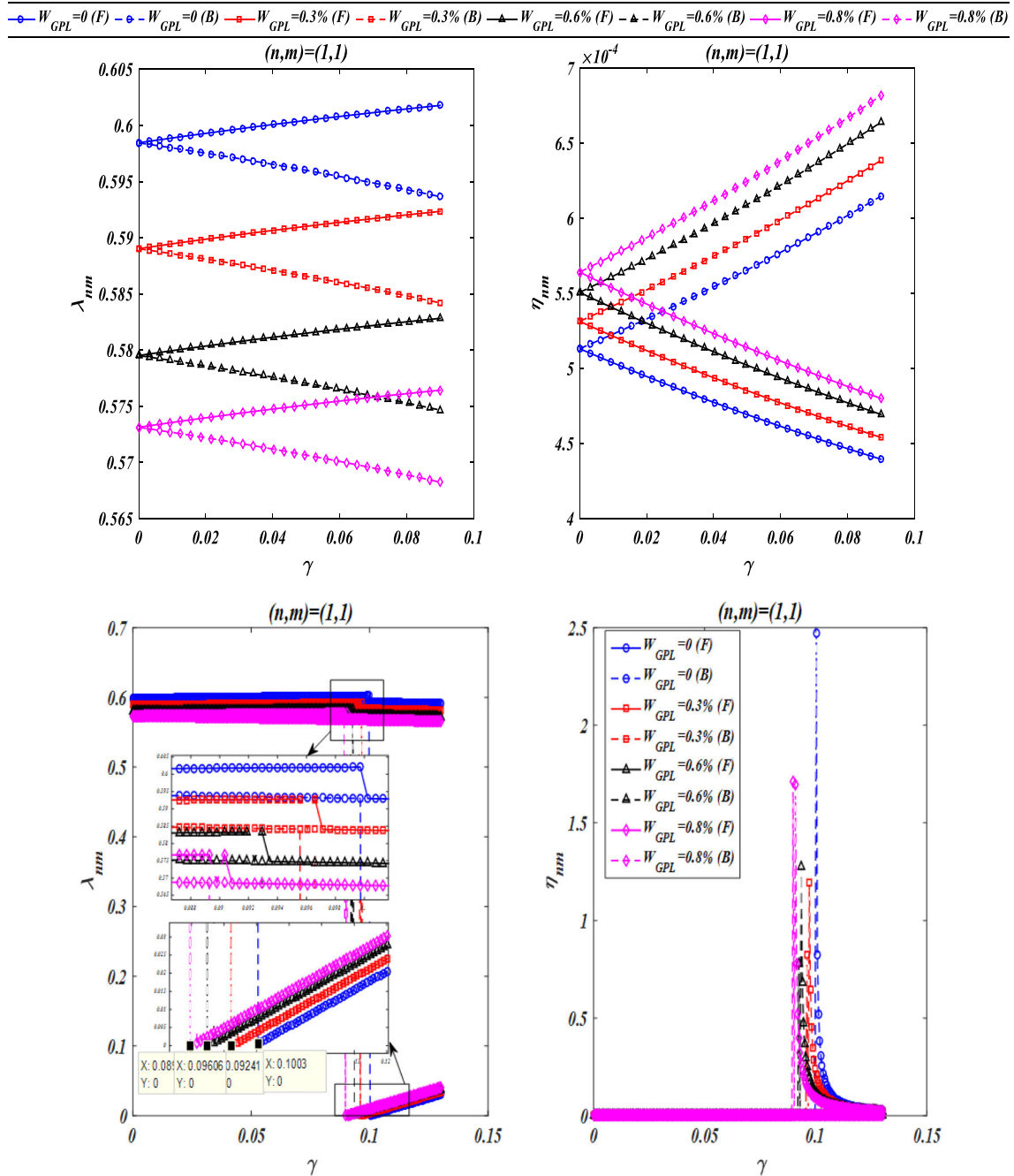


Fig. 10 Effect of volume fraction of GPLs on loss factor and frequency of a rotating nanocomposite sandwich shell

the composite matrix, change the molecular arrangement of the matrix, and increase the force between the molecules. Also, GPLs have the most contact with composite matrix particles due to their flat surface. These reasons increase the rigidity of the structure and, as a result, increase the deflection.

Long rotating structures such as centrifuges are used in industries today. As the aspect ratio increases, the structure becomes thinner and taller; as a result, the strength of the structure decreases. The effect of aspect ratio on the loss factor and frequency of a rotating nanocomposite sandwich shell reinforced by a uniform distribution of GPLs, as well as magnetorheological core, is illustrated in Fig. 11. By increasing the aspect ratio in the rotating shell and considering that it is under axial and rotational load, the frequency of the sandwich shell decreases, and the loss factor of the structure increases, causing the structure to become unstable at lower

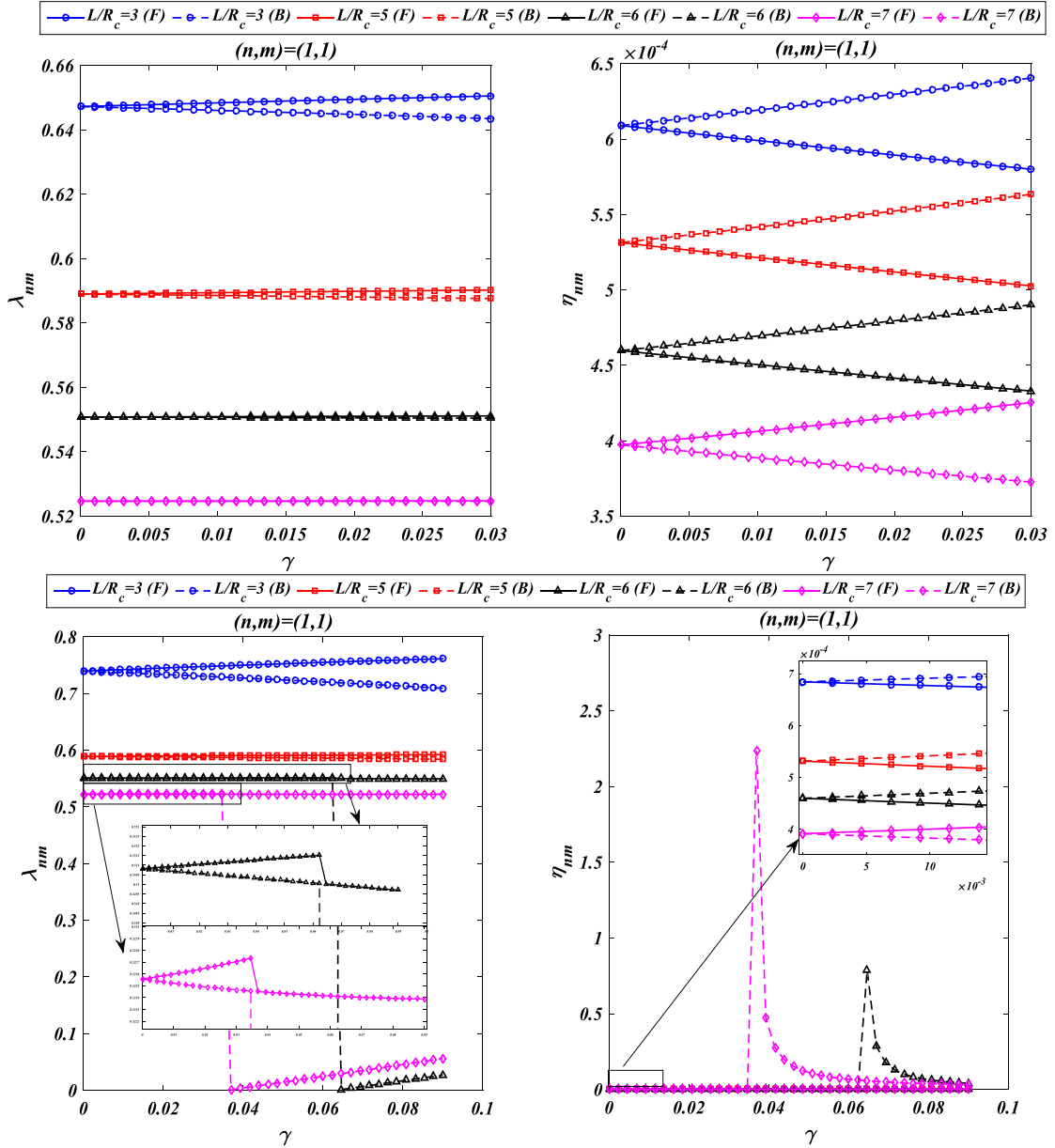


Fig. 11 Effect of length-to-radius ratio on the frequency and loss factor of reinforced rotating nanocomposite sandwich shell

rotational speeds. It can also be seen that the rotational sandwich shell becomes unstable for an aspect ratio of more than six and above with increasing rotational speed, and the backward frequency reaches zero.

6 Conclusion

In this research, the stability of a rotating shell with a magnetorheological core and nanocomposite layers reinforced with GPLs under mechanical and thermal loadings is investigated. One of the things investigated in this research is the use of GPL in the face sheet layers of the sandwich structure and its effect on the stability of the structure. Due to the unique characteristics of GPL, it can be used in the matrix structure to improve the mechanical properties. Also, in this research, different arrangements are considered for the distribution of GPL in the face sheet layers of the sandwich structure. Increasing the volume fraction of GPL in normal conditions increases the stiffness, but when the sandwich structure is under thermomechanical loading, one must be very

careful about adding GPL because it is possible that the properties of the material depend on temperature. Under temperature loading, it causes an increase in thermal axial load and decreases the frequency and stability range of the structure. It is shown that with the increase in the volume fraction of GPL, the frequency of the rotating shell decreases, causing instability in the structure at lower rotational speeds. Increasing the volume fraction of GPLs increases the loss factor of the structure. Considering that the properties of the material are dependent on temperature, increasing the thermal axial load decreases the frequency and stability range of the structure. The lowest frequency of the rotating structure is obtained for the UU distribution, and the highest natural frequency is obtained for the XX distribution. The frequencies of the rotating sandwich shell in AV and VA distributions are almost the same, although the natural frequency obtained in the AV distribution is slightly higher than in VA. Another important case investigated in this research is the use of magnetorheological elastomer as the core of a sandwich structure. In magnetorheological, elastomers play the role of matrix, and magnetic particles act as reinforcement. In the absence of an external magnetic field, the magnetic particles inside the elastomer are randomly distributed, but with the application of an external magnetic field, these particles are aligned with the external field like a chain. When magnetorheological materials are exposed to an external magnetic field, their strength and stiffness change and thus cause a change in the characteristics of the system. The shear modulus and loss factor of the magnetorheological core for the structure are completely dependent on temperature and magnetic field. With the enhancement of the magnetic field, the iron particles in the core are arranged in the direction of the magnetic field, which enhances the stiffness of the core. This point increases the tolerable rotational speed threshold for the rotating sandwich shell. Of course, the optimal value of the magnetic field is in the range of 0.6T, and increasing it more than this value does not have a noticeable effect on the frequency and loss factor for the rotating sandwich shell. In rotating structures, due to gyroscopic effects, the change in rotational speed causes a change in the natural frequencies of the structure. Increasing the rotational speed of the sandwich structure with the elastomeric core reduces the natural frequencies of the structure. As a result, the structure reaches the natural frequency of the structure at a lower rotational speed, which causes the sandwich structure to intensify and become unstable. Temperature and external load affect the stability of the structure. When the frequency of the external excitation force is close to the frequency, it causes resonance. In this research, all the material properties of face sheet layers for the nanocomposite sandwich structure have been considered depending on the temperature. The loss factor and shear modulus of the magnetorheological core are also considered dependent on temperature and magnetic field because the loss factor and shear modulus for core change under temperature and magnetic field. Increasing the temperature in the magnetorheological core causes the destruction of the hard parts of the core. Therefore, increasing the temperature and applying the magnetic field causes the iron particles in the magnetorheological core to find a regular arrangement along the field, and then, the shear modulus is enhanced. As the temperature increases further due to the excessive softness of the elastomer, the shear modulus gradually decreases from its maximum value at the temperature (353 K). The same thing happens with the loss factor. As the temperature increases, the bending stiffness of the face sheet layers for the rotary nanocomposite sandwich shell decreases, and considering that its effect is greater than the increase in the shear modulus of the core due to the increase in temperature, finally, the increase in the temperature of the rotary nanocomposite sandwich shell leads to a decrease in frequency. Then, as a result, it leads to the instability of the structure at higher rotational speeds. When considering the loss factor of the rotating nanocomposite sandwich shell, the behavior is different because the core plays the main role. The loss factor enhances with enhancement of temperature up to (353 K) and then it starts to decrease as the temperature increases and approaches the temperature (333 K). By increasing the axial force on the rotating sandwich shell, the frequency of the structure decreases and causes the rotating shell to reach instability at lower rotational speeds. The enhancement in axial force increases the loss factor of the rotating shell.

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Declarations

Conflict of interest No conflict of interest statement exists in the submission of this article, and all authors approve the article for publication.

Appendix A1

$$\begin{aligned}
N_x^i &= A_{11}^i \frac{\partial}{\partial x} u^i(x, \theta, t) - B_{11}^i \left(\frac{\partial^2}{\partial x^2} w(x, \theta, t) \right) + A_{12}^i \frac{w(x, \theta, t)}{R_i} + A_{12}^i \frac{1}{R_i} \frac{\partial}{\partial \theta} v^i(x, \theta, t) \\
&\quad - B_{12}^i \frac{1}{R_i^2} \left(\frac{\partial^2}{\partial \theta^2} w(x, \theta, t) \right) - N_{Tx}^i \\
M_x^i &= B_{11}^i \frac{\partial}{\partial x} u^i(x, \theta, t) - D_{11}^i \left(\frac{\partial^2}{\partial x^2} w(x, \theta, t) \right) + B_{12}^i \frac{w(x, \theta, t)}{R_i} + B_{12}^i \frac{1}{R_i} \frac{\partial}{\partial \theta} v^i(x, \theta, t) \\
&\quad - \frac{1}{R_i^2} D_{12}^i \left(\frac{\partial^2}{\partial \theta^2} w(x, \theta, t) \right) - M_{Tx}^i \\
N_\theta^i &= A_{12}^i \frac{\partial}{\partial x} u^i(x, \theta, t) - B_{12}^i \left(\frac{\partial^2}{\partial x^2} w(x, \theta, t) \right) + A_{22}^i \frac{w(x, \theta, t)}{R_i} + A_{22}^i \frac{1}{R_i} \frac{\partial}{\partial \theta} v^i(x, \theta, t) \\
&\quad - \frac{1}{R_i^2} B_{22}^i \left(\frac{\partial^2}{\partial \theta^2} w(x, \theta, t) \right) - N_{T\theta}^i \\
M_\theta^i &= B_{12}^i \frac{\partial}{\partial x} u^i(x, \theta, t) - D_{12}^i \left(\frac{\partial^2}{\partial x^2} w(x, \theta, t) \right) + B_{22}^i \frac{w(x, \theta, t)}{R_i} + B_{22}^i \frac{1}{R_i} \frac{\partial}{\partial \theta} v^i(x, \theta, t) \\
&\quad - \frac{1}{R_i^2} D_{22}^i \left(\frac{\partial^2}{\partial \theta^2} w(x, \theta, t) \right) - M_{T\theta}^i \\
N_{x\theta}^i &= A_{33}^i \frac{1}{R_i} \frac{\partial}{\partial \theta} u^i(x, \theta, t) + A_{33}^i \frac{\partial}{\partial x} v^i(x, \theta, t) - 2B_{33}^i \frac{1}{R_i} \left(\frac{\partial^2}{\partial x \partial \theta} w(x, \theta, t) \right) \\
M_{x\theta}^i &= B_{33}^i \frac{1}{R_i} \frac{\partial}{\partial \theta} u^i(x, \theta, t) + B_{33}^i \frac{\partial}{\partial x} v^i(x, \theta, t) - 2D_{33}^i \frac{1}{R_i} \left(\frac{\partial^2}{\partial x \partial \theta} w(x, \theta, t) \right) \\
Q_{xz}^c &= A_{11}^c \left(\frac{u^t(x, \theta, t) - u^b(x, \theta, t)}{h_c} + \frac{1}{2} \frac{(h_t + 2h_c + h_b)}{h_c} \left(\frac{\partial}{\partial x} w(x, \theta, t) \right) \right) \\
Q_{z\theta}^c &= A_{11}^c \left(\frac{1}{R_c} \frac{\partial}{\partial \theta} w(x, \theta, t) - \frac{1}{2R_c} (v^t(x, \theta, t) + v^b(x, \theta, t)) + \frac{1}{4R_c} \left(\frac{h_b}{R_b} - \frac{h_t}{R_t} \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right) \right) \\
&\quad + \frac{1}{h_c} (v^t(x, \theta, t) - v^b(x, \theta, t)) + \frac{1}{2h_c} \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) \left(\frac{\partial}{\partial \theta} w(x, \theta, t) \right)
\end{aligned}$$

Appendix A2

$$\begin{aligned}
\alpha_1 &= \frac{h_t + 2h_c + h_b}{2h_c}, \quad \alpha_2 = \left(\frac{1}{2h_c} \right) \left(\frac{h_b}{R_b} + \frac{h_t}{R_t} \right) + \left(\frac{1}{R_c} + \frac{h_b}{4R_b R_c} - \frac{h_t}{4R_t R_c} \right) \\
\alpha_3 &= \left(\frac{1}{h_c} - \frac{1}{2R_c} \right), \quad \alpha_4 = \left(\frac{1}{h_c} + \frac{1}{2R_c} \right), \quad \alpha_5 = \left(\frac{1}{4R_c^2} - \frac{1}{h_c^2} \right)
\end{aligned}$$

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